OECD Compendium of Productivity Indicators 2025



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Foreword

Productivity growth shapes economic development and underpins improvements in living standards. The pace and pattern of changes in productivity differ widely across countries and over time, reflecting the interplay of a wide range of factors – structural and cyclical, international and domestic. Internationally comparable indicators of productivity are central for assessing cross-country variation in productivity trends and gauging economic performance.

Since its first edition in 2006, the OECD Compendium of Productivity Indicators has compiled a comprehensive set of cross-country statistics on labour productivity, multifactor productivity, and related indicators. It offers an in-depth overview of short- and long-term trends in productivity across OECD countries and, where possible, accession countries, while also examining key components such as capital and labour inputs. In addition, the Compendium presents more granular insights into productivity performance, highlighting differences across industries and between small and medium-sized enterprises and large firms. This edition also includes an examination of productivity growth adjusted for cyclical economic fluctuations.

The report was coordinated by Nhung Luu and completed under the supervision of Jarmila Botev and Annabelle Mourougane, with contributions from Alexander Jaax. Chapters were prepared by Nhung Luu (Chapters 1, 3, 5, 6 and 7), Hector Moreno (Chapters 2 and 4), Tom Arend (Chapter 8), Yann Dorville (Chapter 9). Julia Huf and Gueram Sargsyan provided excellent statistical contributions.

The report was formatted by Virginie Elgrably.

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Executive summary

Productivity growth remained subdued in 2023 and 2024 amid a shifting geopolitical and economic landscape

Following a 0.2% drop in 2022 compared to 2021, labour productivity across all OECD countries rebounded modestly to 0.6% in 2023. In the euro area, labour productivity fell sharply by 0.9% in 2023, marking the steepest decline since the 2008 financial crisis. Experimental estimates suggest that labour productivity growth is likely to have grown modestly at around 0.4% in 2024 on average across OECD countries, excluding Türkiye. While Artificial Intelligence (AI), particularly Generative AI, is expected to positively shape future productivity trends if the right policies are in place, its impact is not yet evident in the productivity statistics.

Evidence for 2023 shows that Multifactor Productivity (MFP) growth, which captures the joint efficiency of labour and capital inputs, either stagnated or declined in most countries, explaining most of the weak labour productivity performance. By contrast, the contribution of capital deepening, while still negative, showed a slight improvement in 2023.

Investment rates varied across OECD countries in 2023. Around half of OECD countries experienced an increase in investment, while the other half saw stagnation or even a decline. This variation likely reflects differences in how firms and households responded to heightened uncertainty, rising costs, and tighter credit conditions.

Productivity performance varied across countries, industries and firms

Aggregate figures mask considerable variation across countries. Around half of OECD countries saw productivity gains in 2023, whereas the other half experienced declines of varying magnitude. Robust labour productivity growth in several non-EU countries helped support the modest increase in the OECD average.

Evidence suggests that cross-country variation in productivity performance was primarily shaped by structural factors, including differences in the business environment and regulatory frameworks, rather than cyclical factors such as labour hoarding.

Labour productivity growth in 2023 was mostly driven by within-industry developments rather than shifts across industries. In about half of the countries with data available, the within-industry contribution was negative or near zero, pulling down overall productivity. Labour productivity growth in manufacturing was the key contributor to economy-wide productivity gains in several countries, including Slovak Republic and Denmark. Energy-related activities, on the contrary, exerted a significant drag on productivity growth in countries like Greece and Croatia. Technology-intensive activities saw a slowdown in productivity growth in many OECD countries.

Large firms also played an important role in shaping productivity developments. On average, they tend to achieve higher labour productivity levels than their smaller counterparts across OECD countries, although this pattern varies by industry. The gap was more pronounced in manufacturing than in business services, reflecting economies of scale and the capital-intensive nature of production.

1. Productivity in a shifting geopolitical and economic landscape

Since its first edition in 2006, the OECD Compendium of Productivity Indicators has shed light on short- and long-term trends in labour and multifactor productivity across OECD countries and where possible accession countries over the period 1995-2023. It has also examined trends in key components of productivity, namely capital and labour inputs. Beyond economy-wide developments, the publication also offers granular insights at industry and firm level. In addition, it provides insights into productivity developments in 2024 and presents two new analyses on multi-factor productivity by industry and on cyclical adjustment of multi-factor productivity in manufacturing.

Weak overall productivity growth in 2023 and 2024

In 2023, labour productivity – measured as GDP per hour worked – for the total economy averaged across all OECD countries saw a modest increase of 0.6%. However, it fell by 0.9% in the euro area – the largest drop since 2009 – continuing a downward trajectory observed since 2021. In contrast, labour productivity increased by 1.6% in the United States, close to the pace seen in 2019 (Figure 1.1).

The subdued labour productivity growth in 2023 unfolded against the backdrop of a challenging macroeconomic environment. The global economy slowed markedly to 1.7% in 2023 from 3% in 2022 amid headwinds, such as tighter financial conditions and heightened geopolitical tensions. Despite modest economic growth, labour markets remained resilient. Hours worked grew by 1.1% on average across OECD countries, mostly through an increase in employment.

At the same time, globalisation appeared to have lost momentum, with global trade growth weakening notably (OECD, 2024[1]). Since trade and foreign direct investments play a vital role in facilitating knowledge diffusion and driving innovation, a weakening of global economic integration is likely to weigh on future productivity performance (Aiyar et al., 2023[2]).

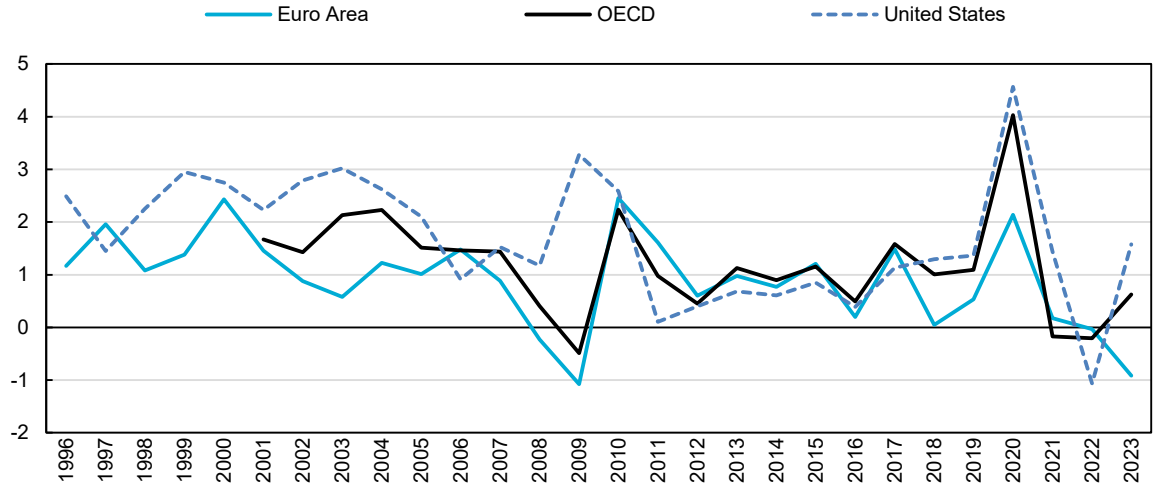
In addition, real interest rates remained elevated in 2023 in most countries (OECD, 2024[1]). Restrictive financial conditions in most major economies, with high lending rates and tight credit standards, continued to dampen firms' investment and economic activity. Elevated borrowing costs tend to raise operational expenses and create uncertainty around the interest rate outlook, making it harder for firms to commit to long-term investments (Duquerroy, Istrefi and Mouabbi, 2024[3]). This, in turn, can slow innovation and impede productivity growth, as constrained access to credit limits firms' capacity to adopt new technologies and invest in efficiency-enhancing measures.

Labour productivity growth is estimated to have stagnated at around 0.4% in 2024 on average across OECD countries excluding Türkiye (Chapter 9). However, these estimates are subject to significant uncertainty. Labour productivity growth is expected to have improved modestly in OECD Asian countries, averaging 1.8% in 2024. In contrast, productivity growth would have remained stagnant in OECD South

American countries and several European economies outside the euro area. Meanwhile, labour productivity growth is expected to have declined slightly in both the euro area and North America.

Figure 1.1. Labour productivity growth since 1995

GDP per hour worked, Per cent



Note: Official data on labour productivity data across OECD countries are only available from 2011 onwards due to the unavailability of data on hours worked for Korea prior to that year, following a methodology revision by Statistics Korea in 2017. Data of this series for years before 2011 are based on estimates using past vintage data.
Source: OECD Productivity Database (2025).

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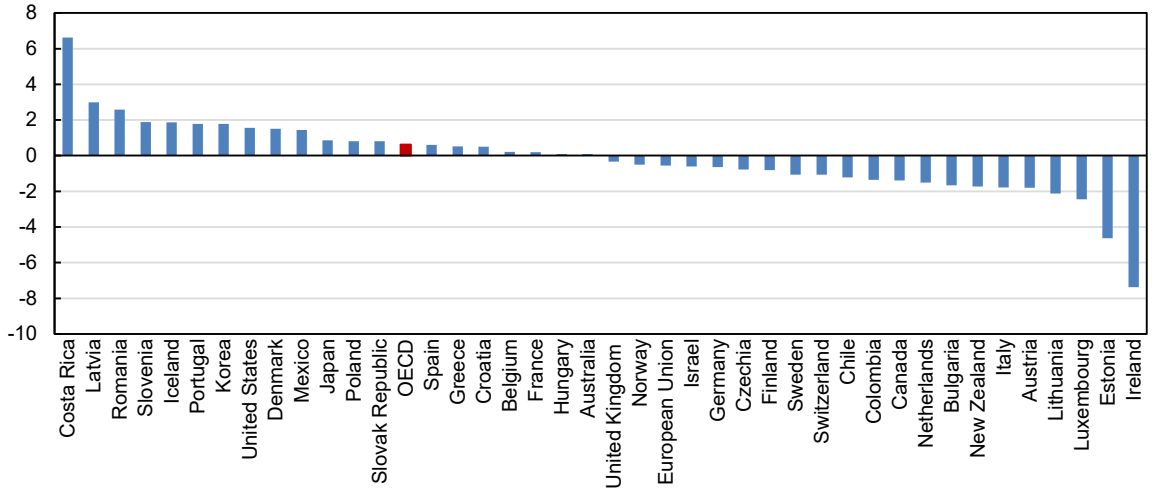
Productivity growth differed across countries

The OECD average labour productivity growth masks significant variation across economies in 2023. It was positive in about half of the countries, and negative in the other half, with varying magnitudes. Strong labour productivity gains were seen in a number of non-EU countries, contributing to the positive growth in the OECD average. Costa Rica, Latvia and Romania were the top performers, whereas Ireland, Estonia, and Luxembourg experienced the steepest declines (Figure 1.2).

In 2023, economic activity in the euro area, and particularly Central and Eastern European countries, was notably weak, with GDP growth slowing to 0.4% from 3.5% in 2022, reflecting high energy prices and other headwinds (Chapter 2). On the contrary, growth remained robust in the United States and Japan experienced above-trend growth. Although the overall economy slowed down in the euro area, employment rates stayed robust, and total hours worked in this region rose by 1.6% in 2023. Labour hoarding, where firms retain employees despite weak demand, contributed to a decline in labour productivity in many countries (ECB, 2024[4]).

Figure 1.2. Labour productivity growth across countries in 2023

GDP per hour worked, Total economy, Per cent



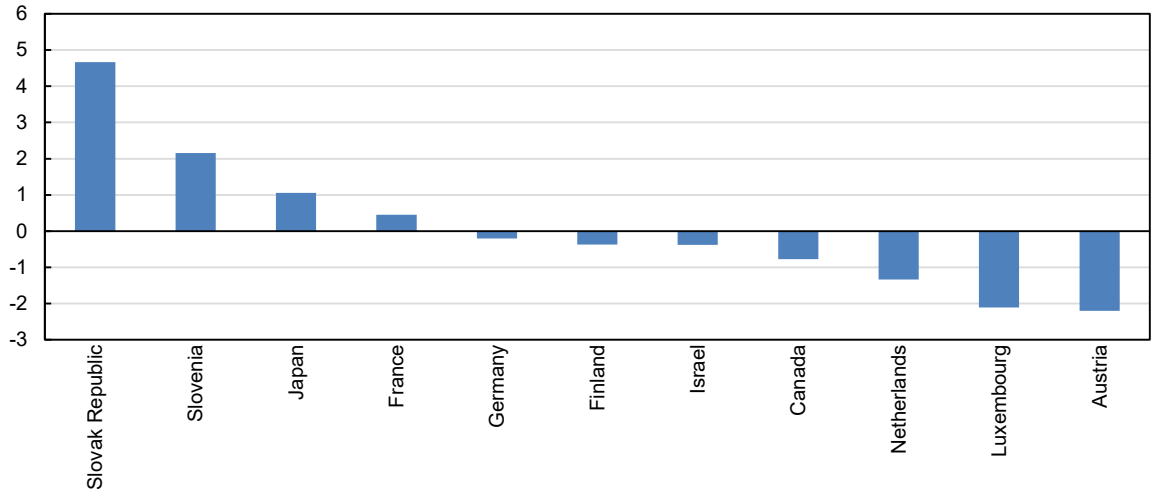
Source: OECD Productivity Database (2025).

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As an alternative measure of productivity, multifactor productivity (MFP) growth captures changes in economic performance beyond those explained by capital and labour. MFP growth either stagnated or even turned negative in most OECD countries in 2023 (Figure 1.3). The slowdown was particularly pronounced in Luxembourg and Austria, where MFP fell by around -2% in 2023. By contrast, MFP grew fast in Slovak Republic and Slovenia, with growth rates of 4.6% and 2%, respectively in the same period.

Figure 1.3. Multifactor productivity growth in 2023

Total economy, Per cent



Source: OECD Productivity Database (2025).

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Slow MFP growth explained most of the weak labour productivity performance in 2023, with capital deepening playing only minor role.

The slowdown in MFP growth was the main factor behind the weak labour productivity performance. The dominance of MFP contribution over capital deepening, defined as capital services per hour worked, in explaining labour productivity growth has been evident in the past for long (OECD/APO, 2022[5]). While the contribution of capital deepening was negligible in some countries, it was negative in many others at the total economy level (Chapter 2).

This variation in the role of capital deepening across countries in 2023 partly reflects differences in investment patterns – an essential component in measuring capital services. Investment rates (i.e. investment over GDP) varied across OECD countries in this period. Around half of the OECD countries experienced an increase in investment rates, while the other half saw stagnation or even a decline. This variation likely reflects differences in how firms and households responded to heightened uncertainty, rising costs, and tighter credit conditions.

Investment rates also varied across asset types. Much of the increase was concentrated in construction-related assets, such as other buildings and structures, and in non-ICT capital goods, including other machinery, equipment, and weapons systems. By contrast, investment in ICT equipment and intellectual property products (IPP), key enablers of innovation and digital transformation, declined in most countries (Chapter 4).

Over time, capital deepening has shown a declining contribution to labour productivity growth (Chapter 6). This can be attributed in part to a slowdown in net investment since the global financial crisis. The overall weakness in investment is likely driven by subdued spending on non-digital tangible assets, coupled with rising depreciation (OECD, 2025[6]).

Labour productivity growth continues to be shaped by within-industry developments and large firms' productivity growth

In most OECD countries, labour productivity growth in 2023 was primarily shaped by changes within industries. The cross-industry reallocation effect, reflecting shifts in total hours worked between industries, played a minimal role in the same year (Figure 1.4). In about half of the countries with available data, the within-industry contribution was negative or near zero, pulling down overall productivity growth (Chapter 3).

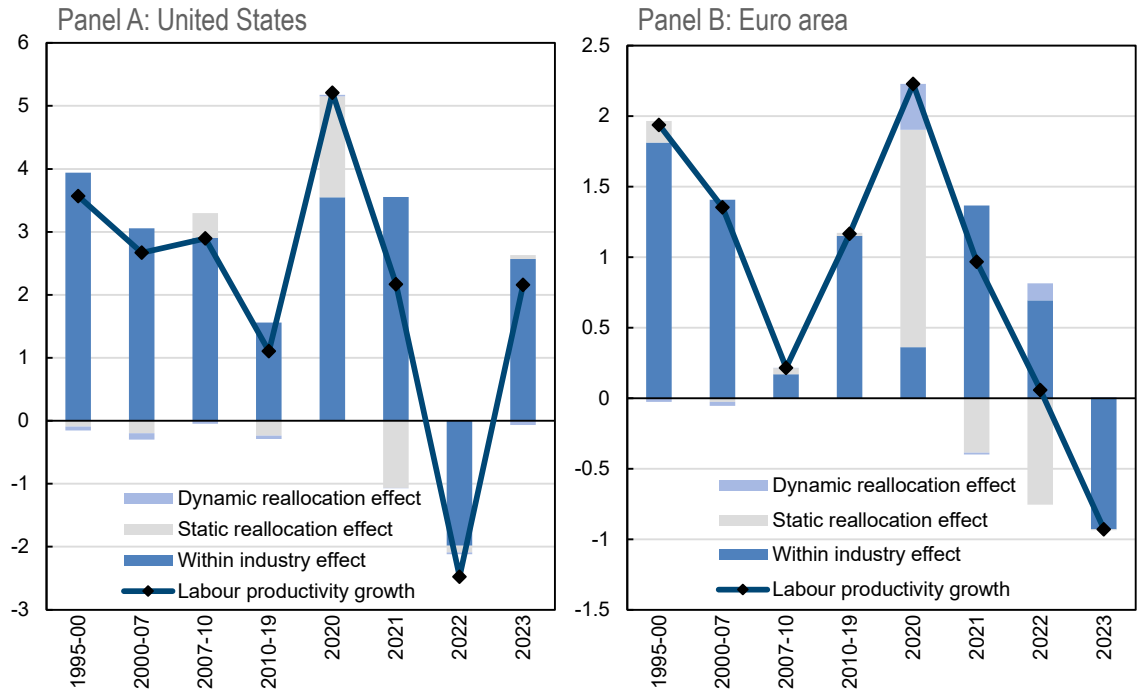
Knowledge-intensive industries, including information and communication and finance, recorded negative productivity growth in many countries in 2023. Manufacturing made the strongest positive contribution to overall labour productivity growth in several countries, including Slovak Republic and Denmark. Meanwhile, energy-related activities, comprising electricity, gas, steam and air conditioning supply, had a strong negative impact on economy-wide labour productivity growth in Greece and Croatia (Chapter 3).

While the overall impact of cross-industry reallocation was limited, 2023 saw a relative increase in the share of hours worked in low-productivity sectors, such as transportation, accommodation, and food services (see Chapter 3). This shift partly reflects the easing of labour shortages, which had persisted since the COVID crisis. In several OECD countries, quit rates slowed and job vacancies decreased in 2023 (OECD, 2024[1]).

Large firms in OECD countries consistently exhibit higher labour productivity on average compared to their smaller counterparts (Chapter 7). Indeed, large firms (those with more than 250 employees) tend to invest more heavily in information and communication technology (ICT) capital and digitalisation and account for a signification share of innovation (Garcia-Masia, Hsieh and Klenow, 2019[7]; Lopez-Garcia et al., 2024[8]) and trade more than smaller firms. In manufacturing, the productivity gap between large and smaller firms is on average more pronounced than in the business economy as a whole, reflecting returns to scale in capital-intensive activities.

Figure 1.4. Contributions to labour productivity growth

Percentage points



Note: The static reallocation effect refers to the reallocations of hours worked between industries with different productivity levels. The dynamic reallocation effect accounts for the reallocations of hours worked between industries with different productivity growth rates. The sum of the two makes up the cross-industry reallocation effect. The within-industry effect reflects labour productivity developments that are not the result of reallocation of hours worked between industries. It is measured by the labour productivity growth in each industry weighted by the industry share in total value added (Chapter 5). For United States, these three effects do not fully sum to the labour productivity growth due to the use of Fisher volume indices for GVA. However, the residual is typically small, especially when averaged over several years (see the methodological note).
Source: Authors' calculation based on OECD Productivity Database (2025).

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Cross-country variation in productivity growth primarily reflected structural factors in 2023

Divergent trends in labour productivity growth are usually shaped by a combination of factors, including short-term cyclical factors, i.e. differences in the position of individual economies within the economic cycle, and structural aspects. In many euro area countries, both labour productivity and MFP growth tend to be highly procyclical, being low in times of low output growth and high when the economy rebounds. In contrast, countries such as the United States and Australia exhibit less procyclical behaviour, as labour is more flexibly adjusted to production (Dossche, Gazzani and Lewis, 2023[9]).

Yet, evidence suggests that cyclical factors had limited effects on productivity developments across OECD countries in 2023, at least in manufacturing (Chapter 7). Instead, variation in productivity performance was primarily shaped by structural factors, including differences in the business environment and regulatory framework.

Challenges and opportunities for future productivity trends

Looking forward, productivity growth will be shaped by both ongoing challenges and transformative opportunities. On the one hand, geopolitical tensions and economic uncertainty appear likely to reduce investment as firms delay or scale back investments, leading to slower technological innovation and adoption and decreased overall efficiency in production processes.

On the other hand, new advances in Artificial Intelligence (AI), particularly Generative AI, could offer promising avenues for boosting innovation and driving future productivity growth. Yet, the potential productivity gains from AI come with notable challenges. Widespread adoption of AI will require new skillsets, placing pressure on both workers and employers to adapt. Unresolved issues around intellectual property and data ownership further complicate the regulatory landscape and may hinder innovation and diffusion (OECD, 2023[10]). Beyond economics, AI deployment also presents broader social, environmental and national security challenges (OECD, 2022[11]; OECD, 2024[12]). The energy-intensive nature of AI computing can lead to increased greenhouse gas emissions, potentially offsetting some of the productivity gains (OECD, 2023[13]). Reaping the benefit of AI will require complementary investments and institutional adaptation and productivity gains may only become visible over time (see Box 1.1).

Box 1.1. AI emergence and its implications for productivity

Although Artificial Intelligence (AI) has been studied since the 1950s, only in recent years has it begun to transform economies and societies at scale. This shift has been driven by major advances in computing power, machine learning techniques, and the availability of large datasets. The emergence of generative AI, particularly following the release of tools like ChatGPT and Gemini in early 2022, has further accelerated the integration of AI across many areas of work and daily life.

There is no universally-agreed definition of AI

Different definitions of AI have been adopted across communities, commonly addressing AI as technologies that perform tasks by mimicking human-like senses, learning, and actions. For instance, OECD adopted a shared definition of AI, which is used in policymaking and legislative work (OECD, 2024[12]). According to this definition, AI refers to machine-based systems that, for explicit or implicit objectives, process input data and generate outputs, such as predictions, content, recommendations, or decisions that can influence physical or virtual environments. These systems require intangible inputs like software, data, and human expertise with substantial computing capacity and complementary technologies. The statistical definition of AI, introduced in the System of National Account 2025, is different and defines AI as “a computer program operating a system capable of recognition, reasoning, communication, and prediction, simulating human recognition, reasoning, and communication”.

Quantifying the impact of macro-economic AI in the long run is challenging

Measuring the impact of AI on productivity presents significant challenges, particularly at the macroeconomic level. Firstly, AI amplifies traditional measurement challenges such as capturing intangible and public sector production in productivity measures. Without progress in these areas, assessing the effects of AI-related policies will remain difficult. Secondly, while micro-level studies or sector-specific analyses point to some evidence of productivity gains, those gains are harder to identify at the macroeconomic level. The complexity is compounded by the intangible nature of AI investments, such as data, knowledge, skills, or computing power and capacity, which are poorly captured in the current national accounts. Additionally, much like previous technological transformations, such as the rise of the internet, AI’s broader economic benefits often depend on complementary investments and institutional adaptation and may only become visible over time. This delayed impact, coupled with the

diversity in how AI systems are designed and deployed, adds further complexity to measurement. Finally, current research also focuses mainly on advanced economies, neglecting international spillovers.

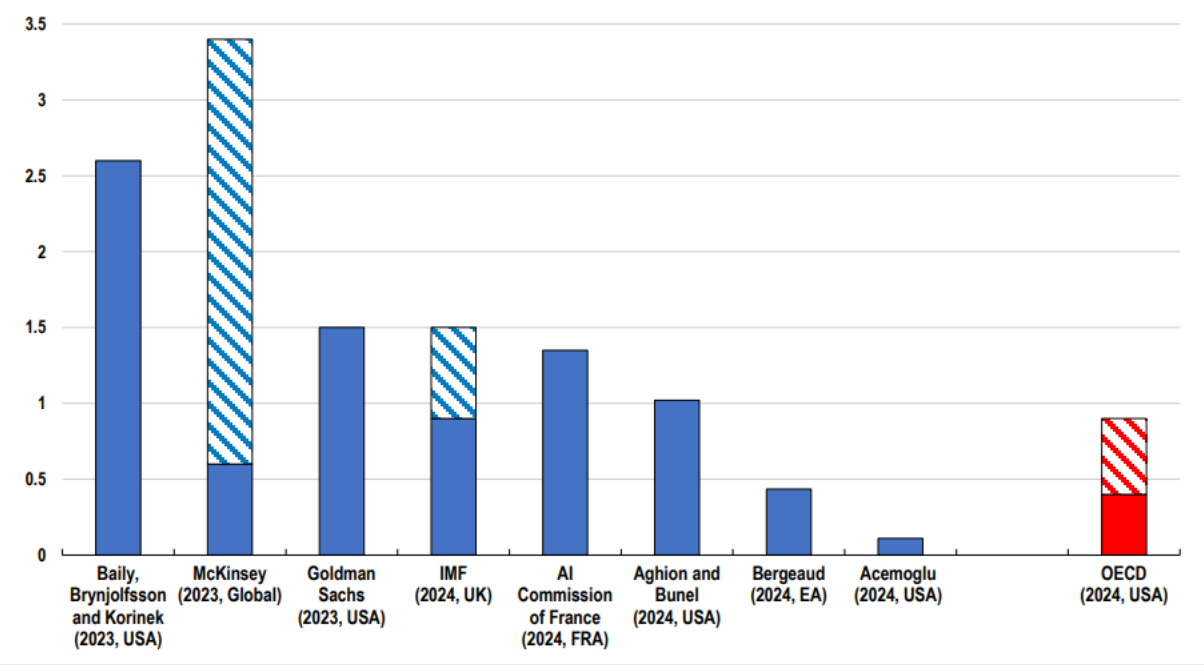
Early estimates of AI's impact on productivity

Estimates of AI's contribution to aggregate productivity growth vary, depending on the modelling framework and underlying assumptions. AI is estimated to raise labour productivity growth by between 0.5 and 3.5 percentage points per year over a ten-year horizon (Figure 1.5). Based on a micro-macro framework which combines micro-level performance gains estimates with exposure of activities to AI and likely future adoption rates, recent OECD modelling suggests that AI's contribution to productivity growth in the G7 will range from 0.2 to 1.3 percentage points, with considerable diversity across countries (Filippucci et al., 2025, forthcoming[14]).

Figure 1.5. AI's predicted annual labour productivity gains over the next ten years across studies

Percentage points

Euro area, the United States, the United Kingdom, and France



Note: The solid bar refers to the lower bound while the striped bar corresponds to the upper bound of the estimate. In cases where modelling predictions primarily focus on TFP, labour productivity is obtained using simple assumptions about the aggregate capital multiplier (Acemoglu, 2024[15]; Aghion and Bunel, 2024[16]; Bergeaud, 2024[17]). The estimates refer to the countries shown in brackets.
Source: (Filippucci, Gal and Schief, 2024[18]) (Figure A.10.)

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OECD/APO (2022), *Identifying the Main Drivers of Productivity Growth: A Literature Review*, OECD Publishing, Paris, https://doi.org/10.1787/00435b80-en. [5]

2. Productivity and economic growth

Context and overview

Economic growth slowed markedly in 2023, both globally and among OECD countries. GDP growth in OECD countries amounted to 1.7% in 2023 after 3.0% in 2022 (OECD, 2024[1]). This deceleration reflected a combination of factors, including the delayed effects of monetary tightening, persistently high inflation, weak global trade, and continued uncertainty amid overlapping global crises (Chapter 1).

Productivity gains reflect the ability to produce more output by combining inputs more efficiently driven by new ideas, technological breakthroughs, and improvements in business models. These innovations transform the production of goods and services, fostering economic growth and ultimately improving living standards and well-being.

To understand these productivity gains in the context of slowing growth, this chapter adopts three complementary analytical perspectives to examine the drivers of economic performance. It begins by analysing labour productivity, breaking it down into the contribution of capital deepening (the increased use of capital per hour worked) and multifactor productivity (MFP), which captures how efficiently capital and labour are combined. This decomposition helps determine whether productivity improvements stem from greater input intensity or from more efficient use of existing inputs.

The chapter then applies a growth accounting framework to break down GDP growth into contributions from labour and capital inputs, along with MFP, which serves here as an indicator of technological progress and innovation. Finally, the chapter examines the role of three labour-related inputs in economic growth: employment, average hours worked, and labour productivity. All in all, these three perspectives provide a comprehensive framework for analysing the relative contributions of input accumulation and efficiency gains to economic growth.

Key findings

- **GDP growth in 2023 was predominantly driven by increases in labour inputs, with limited support from capital deepening or productivity improvements**. Neither increased capital intensity nor efficiency gains were sufficient to raise output per hour worked in 2023, highlighting the persistence of deeper structural barriers to labour productivity growth. This pattern resonates with OECD analyses that highlight subdued investment and slow diffusion of innovation across firms as structural challenges undermining medium-term growth prospects (OECD, 2024[1]).
- **Compared with other labour-related elements such as hours worked and labour productivity, employment expansion played the most significant role in boosting GDP growth in 2023.** Employment levels improved, or remained steady, in most OECD countries.
- **Multifactor productivity (MFP) made a negative or negligible contribution to GDP growth in 2023.** The contribution of capital deepening was also small or negative in most countries.

GDP growth: Contributions from labour, capital and multifactor productivity

Economic growth can result either from increased use of labour and capital inputs or from a more efficient use of these inputs through gains in multifactor productivity (MFP). This section applies a growth accounting framework to decompose growth into changes in three components: labour inputs, capital inputs, and MFP. This approach offers a valuable lens through which to examine the fundamental drivers of economic performance. The method calculates the contribution of labour and capital to GDP growth by weighting their growth rates according to their shares in total production costs.

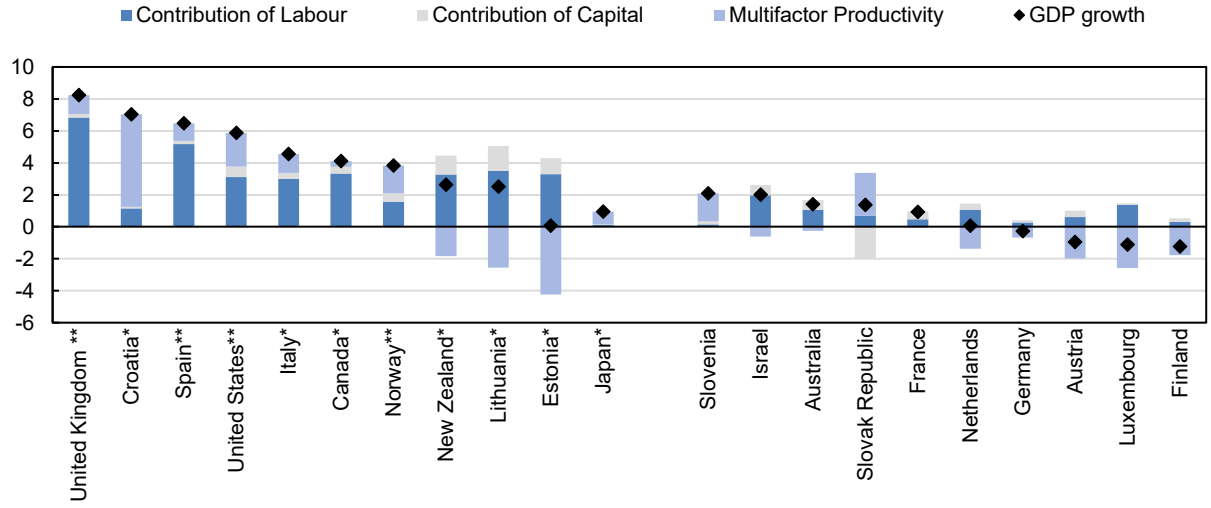
Key findings

- **In the context of slower economic growth in 2023, labour input, measured as total hours worked, remained the main driver of GDP growth across most countries.** Its contribution was generally positive, supporting growth in nearly all countries.
- **In 2023, the contribution of capital stock to GDP growth was more modest than that of labour.** The contribution of capital remained positive in all countries except the Slovak Republic, which was the only growing economy with a negative contribution from capital.
- **The contribution of multifactor productivity (MFP) was negative in 2023 across most countries.** This contribution even turned negative compared with the previous year as in Austria Germany and the Netherlands.

Indicators

Figure 2.1. Labour, capital, and multifactor productivity contributions to GDP growth

Percentage points, 2023 unless otherwise indicated: (*) denotes 2021, (**) denotes 2022



Source: OECD Productivity Database (2025).

StatLink https://stat.link/nltfd0

How to read the indicators

For productivity analysis, the appropriate measure of capital input is the flow of capital services, i.e. the flow of productive services that can be drawn from the productive capital stock. This productive capital stock is the cumulative stock of past investments in capital assets adjusted for losses in their productive capacity (or efficiency) and retirement (Schreyer, Bignon and Dupont, 2003[2]). Conceptually, capital services should not be confused with the value of capital that is measured by the net wealth capital stock. For example, the capital services provided by a taxi relate to the number of trips, distance driven, and comfort of the taxi, rather than the market value of the vehicle, which would instead relate to the net wealth capital stock concept. These services are estimated using the rate of change of the productive capital stock of different capital goods and aggregated using rental prices or user costs shares as weights (as opposed to market price shares used to aggregate net wealth capital stocks).

As part of the revamp of the OECD Productivity database, the method for measuring capital services has been refined. The new estimates of capital services offer a more detailed breakdown by asset type and industry and are fully aligned with national accounts. This consistency stems from the depreciation rates used in calculating asset user costs and return rates are fully in line with those applied by national statistics offices in deriving net capital stock. In addition, this way of calculating capital services does not require the extension of gross fixed capital formation (GFCF) series or strong assumptions to estimate initial capital stocks. This differs from the previous version of the database, which assumed an identical depreciation rate for a given asset in the same industry across countries (OECD, 2025[3]).

MFP growth is measured as a residual, i.e. by the part of GDP growth that cannot be explained by the contributions of labour and capital inputs to GDP growth. Traditionally, MFP growth is seen as a measure of technical change but, in fact, technical change can also be embodied in factor inputs, e.g. improvements in design and quality between two vintages of the same capital asset. In practice, MFP only captures disembodied technical change, e.g. network effects or spillovers from production factors, the effects of better management practices, organisational change and improvements in the knowledge base. Moreover, MFP picks up other factors such as adjustment costs, economies of scale, effects from imperfect competition, variations in capacity utilisation (if not captured by the capital input measures), and errors in the measurement of output, inputs and input weights. For instance, increases in educational attainment or a shift towards a more skill-intensive production process, if not captured by labour input measures (i.e. labour services) will end up in measured MFP. Moreover, tax policies can also affect the measurement of MFP. Profit shifting driven by differences in corporate tax regimes may increase reported output in low tax countries without changes in real production, resulting in inflated MFP levels (Bricongne, Delpeuch and Lopez-Forero, 2023[4]). Therefore, a reliable measure of MFP requires accurate data on both outputs and inputs .

In Figure 2.1, national accounts figure on hours worked for Austria, Estonia, Finland, Greece, Latvia, Lithuania, Poland, Portugal, Sweden and the United Kingdom have been replaced with estimates obtained with the OECD simplified component method described in the Section on Hours Worked and Employment in Chapter 5. However, the impact of this correction on labour productivity growth rates is marginal (Ward, Zinni and Pascal, 2018[5]).

For further methodological information, consult the *OECD Productivity Statistics – Methodological notes* at www.oecd.org/sdd/productivity-stats/OECD-Productivity-Statistics-Methodological-note.pdf and the update of the sources and methodologies used to construct the new OECD Productivity Database (OECD, 2025[3]).

Labour productivity growth: Contributions from capital and multifactor productivity

Labour productivity growth reflects changes in the volume of output for a given volume of hours worked. Higher levels of labour productivity can be achieved if capital is more intensively used in production, and if labour and capital are used together more efficiently, which means higher multifactor productivity growth (MFP). By reformulating the growth accounting framework described in the previous section, labour productivity growth can be decomposed into the contribution of capital and MFP.

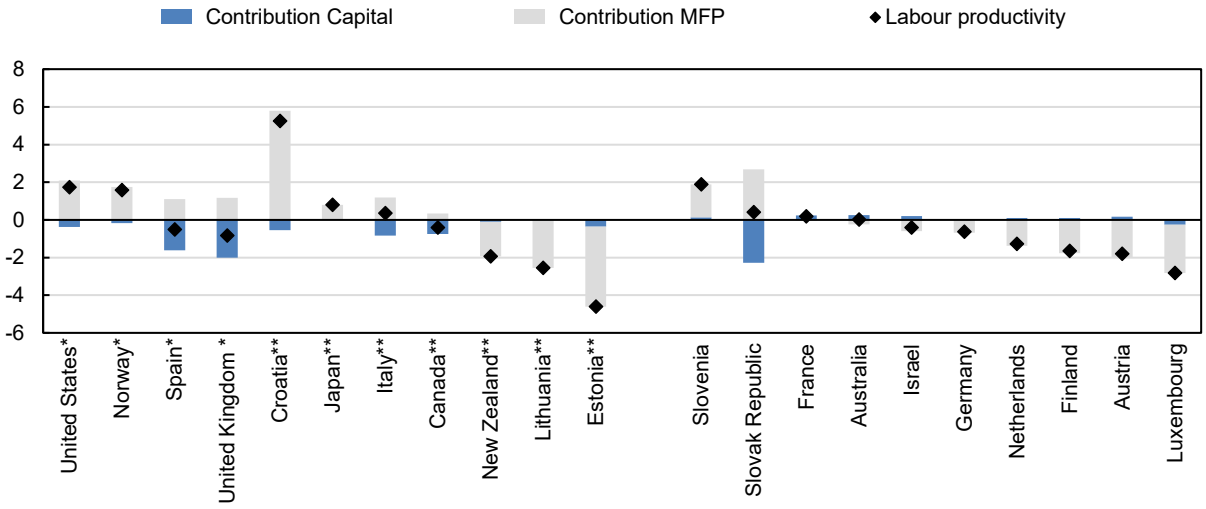
Key findings

- **In 2023, labour productivity growth was modest or negative in most countries, as both capital deepening and multifactor productivity (MFP) made limited or negative contributions.** In a few cases, these two components offset each other, but more often moved in the same direction, jointly pulling labour productivity into negative territory. In a context of slower economic activity, this pattern suggests that neither increased capital intensity nor efficiency gains alone seem sufficient to raise output per hour worked.
- **In 2023, MFP growth was the main contributor to changes in labour productivity, although its influence was predominantly negative.** The largest negative contributions were observed in Luxembourg and Austria.
- **The contribution MFP growth shows a high degree of persistence.** Most countries have exhibited a persistent trend of negative MFP contributions over the past two years, in line with findings from the 2024 edition of the Compendium (OECD, 2024[6]).
- **In 2023, the contribution of capital deepening to labour productivity growth was small or negative in most countries.** Only in Australia the capital intensity offset the negative impact of MFP, resulting in near-zero labour productivity growth, but this is not a systematic pattern since the Slovak Republic, for instance, experienced the opposite scenario where changes in capital input dampened the positive effects of MFP gains.
- **The contribution of capital deepening remained relatively stable over the past years, with most countries showing consistently low or slightly negative contributions.** Substantial increases in capital per worker remain rare.

Indicators

Figure 2.2. Contributions to annual labour productivity growth: Capital stock to output ratio, capital quality and multifactor productivity

Percentage points, 2023 unless otherwise indicated: (*) denotes 2021, (**) denotes 2022



Source: OECD Productivity Database (2025).

StatLink https://stat.link/jal215

How to read the indicators

As explained in the previous section, capital services are calculated using the refined methodology underlying the OECD Productivity Database (2025[3]). This approach aligns fully with national accounts and delivers a more granular breakdown by asset type and industry. One key advantage of this revised method is that it computes capital services directly, without constructing estimates of productive capital stock. However, this simplification comes at the cost of losing the ability to decompose the contribution of capital services to labour productivity growth into the separate effects of capital stock and capital quality.

While MFP growth contributes to both GDP and labour productivity growth, the corresponding contributions are different. MFP growth contributes to GDP growth with a weight equal to one, but it is weighted by the inverse of the labour cost share to calculate its contribution to labour productivity growth.

In Figure 2.2, national accounts data on hours worked for Austria, Estonia, Finland, Greece, Latvia, Lithuania, Poland, Portugal, Sweden and the United Kingdom have been replaced with estimates obtained with the OECD simplified component method described in the Section on Hours Worked and Employment in Chapter 5. However, the impact of this correction on labour productivity growth rates is marginal (Ward, Zinni and Pascal, 2018[5]).

For further methodological information, consult the *OECD Productivity Statistics – Methodological notes* at www.oecd.org/sdd/productivity-stats/OECD-Productivity-Statistics-Methodological-note.pdf and the update of the sources and methodologies used to construct the new OECD Productivity Database (OECD, 2025[3]).

GDP growth: Contributions from employment, hours worked per worker, and labour productivity

This section applies a growth decomposition to quantify the contributions of three labour-related inputs: employment, hours worked, and labour productivity. The approach helps distinguish whether growth is primarily driven by more intensive use of labour (changes in employment or longer working hours) or by gains in efficiency (i.e. changes in labour productivity). Overall, this approach aims at offering alternative insights into recent macroeconomic developments.

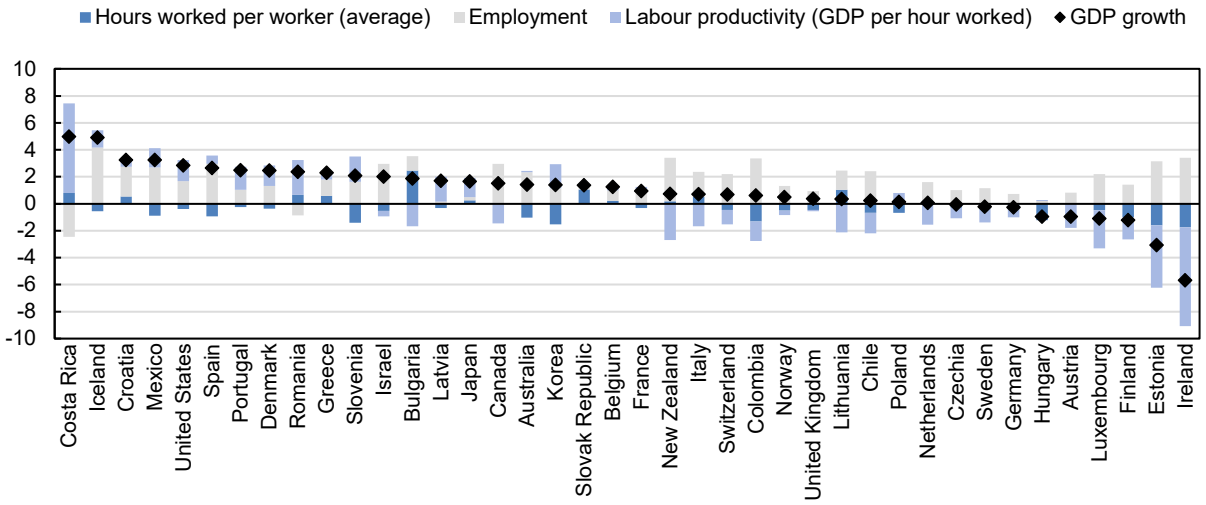
Key findings

- **In 2023, changes in employment, measured as the number of people employed, were often the largest contributor to economic growth across OECD countries –** in particular when compared to changes in other labour-related components such as hours worked and labour productivity. In most countries, the contributions of these three components displayed divergent trends, often offsetting one another's impact on growth. However, in five countries all these three factors jointly aligned to support GDP growth: Croatia, Greece, Japan, the Slovak Republic and Belgium.
- **In 2023, labour productivity growth acted as a key determinant of economic performance, boosting growth in high-performing economies and holding it back in those with weak or declining output.** Nearly half of OECD countries recorded a negative contribution from labour productivity, particularly those with the largest GDP reductions, including Ireland, Estonia, Finland, and Luxembourg. At the same time, productivity gains were the leading contributor to growth in many of the fast-growing economies, such as Costa Rica, Romania, Slovenia, and Korea.
- **Of the three labour-related components, changes in hours worked contributed the least to GDP growth in 2023.** It generally had a small but negative impact in roughly three-quarters of OECD countries, though it played a notable role in supporting growth in Bulgaria and dampening it in Korea.

Indicators

Figure 2.3. Contributions to annual GDP growth from changes in labour productivity, hours worked, and employment in 2023

Percentage points



Source: OECD Productivity Database (2025).

StatLink https://stat.link/om1742

How to read the indicators

In Figure 2.3, statistics on hours worked for Austria, Estonia, Finland, Greece, Latvia, Lithuania, Poland, Portugal, Sweden and the United Kingdom have been replaced with estimates obtained with the OECD simplified component method described in the Section on Hours Worked and Employment in Chapter 5. However, the impact of this correction on labour productivity growth rates is marginal (Ward, Zinni and Pascal, 2018[5]).

For further methodological information, consult the *OECD Productivity Statistics – Methodological notes* at www.oecd.org/sdd/productivity-stats/OECD-Productivity-Statistics-Methodological-note.pdf and the update of the sources and methodologies used to construct the new OECD Productivity Database (OECD, 2025[3]).

Data sources

OECD Economic Outlook: Statistics and Projections (database), http://data-explorer.oecd.org/s/1v5.

OECD Employment and Labour Market Statistics (database), http://data-explorer.oecd.org/s/1v6.

OECD National Accounts Statistics (database), http://data-explorer.oecd.org/s/1v7.

OECD Productivity Statistics (database), http://data-explorer.oecd.org/s/1v8.

OECD Productivity Dashboard, www.oecd.org/en/data/dashboards/oecd-dashboard-of-productivity-indicators.html.

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OECD (2024), *OECD Economic Outlook, Volume 2024 Issue 2*, OECD Publishing, Paris, https://doi.org/10.1787/d8814e8b-en. [1]

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3. Investment

Context and key findings

Disaggregating investment and capital by asset type provides valuable insights into the key drivers of GDP and productivity growth. This approach makes it possible to evaluate developments in specific areas, such as the extent of investment in Information and Communication Technology (ICT). Moreover, each asset type contributes differently to economic performance, influencing both GDP and productivity in distinct ways.

In an increasingly digital economy, the most valuable firms – across high-tech, pharmaceutical, automotive, and financial services industries – derive a large share of their competitiveness and market value from intangible rather than physical capital. The growing adoption of Artificial Intelligence (AI) further underscores the importance of investments in intangible assets, such as data, marketing, and organisational capital. Despite their critical role in driving innovation and competitiveness, their coverage in national accounts remains limited. However, the 2025 revision of the System of National Accounts will include investments in new types of intangibles, such as data assets (SNA, 2025[1]).

Key findings

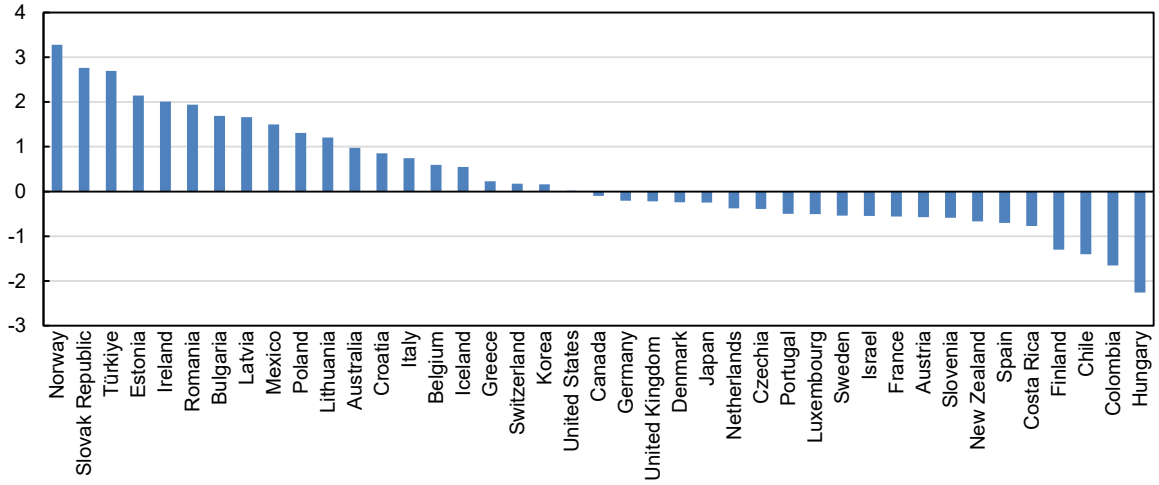
- **In about half of the OECD countries, the investment rate, measured as gross fixed capital formation (GFCF) as a share of GDP, grew in 2023**, with the most significant increases observed in Norway, Slovak Republic and Türkiye. In contrast, Hungary and Colombia experienced the largest declines, with investment rates in 2023 dropping by two or more percentage points. Among the G7 countries, investment rates remained broadly stable, except for France where they declined by around 0.5 percentage points (Figure 3.1).
- **Dwellings and other buildings and structures made up the largest portion of total investment in 2023, exceeding 50% in almost two thirds of the countries** (Figure 3.2). However, patterns differed across OECD countries: investment in Dwellings as a share of GDP fell in most cases compared to 2022, whereas investment in Other buildings and structures increased in the majority of member countries (Figure 3.3).
- **Intellectual Property Products (IPP) in 2023 also accounted for a significant portion of investment in several countries, contributing to more than a third of total capital formation in Ireland, Switzerland and the United States.** By contrast, investment in IT equipment constituted the smallest share in most countries (Figure 3.2). However, investment in both IT equipment and IPP decreased as a proportion of GDP over the same year (Figure 3.3).
- **Investment in Other machinery, equipment, and weapons played a significant role in driving investment rate growth in several countries, such as Bulgaria, and Slovak Republic and accounted for the largest share of total investment among both SMEs and large firms in both countries** (EIB, 2023[2]; EIB, 2023[3]). In Costa Rica and Latvia, a decline in investment relative to GDP in this category was the main factor behind the overall reduction in investment rate growth (Figure 3.3).

- **Over half of the OECD countries experienced a decline in the investment rate in ICT assets in 2023 and around half of the countries reported reduced investment in intangible assets**. The largest declines in ICT asset investment were observed in Mexico and New Zealand, while the largest increases were recorded in Slovak Republic and Czechia. For intangible assets, Ireland and Israel reported the largest increases, whereas Slovenia and Denmark experienced the largest declines. In most countries, however, the magnitude of changes – be it an increase or decrease – was relatively smaller for ICT and intangible assets compared to tangible and non-ICT assets, partly reflecting the dominant share of the latter in the overall asset composition (Figure 3.4).

Indicators

Figure 3.1. Annual change in investment rate

2023, Total economy, Percentage points

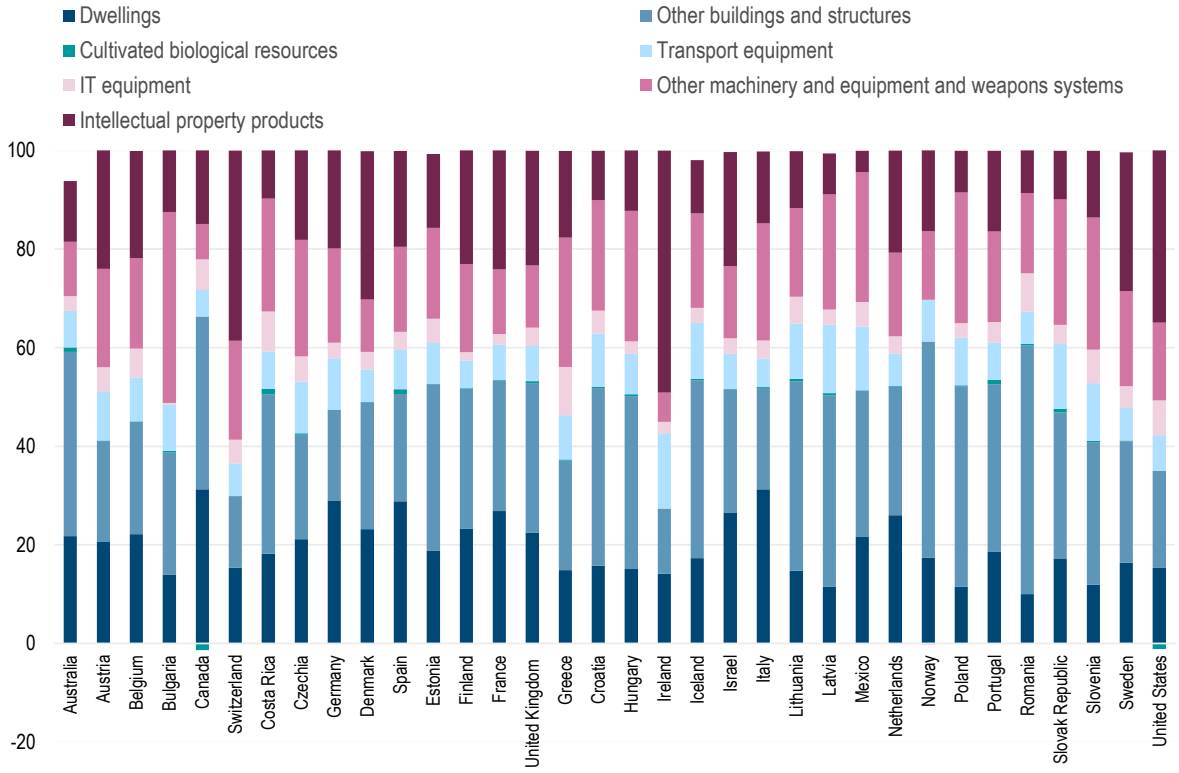


Source: OECD Productivity Database (2025).

StatLink https://stat.link/4mb0gs

Figure 3.2. Decomposition of investment by assets

2023, Total economy, Percentage of total GFCF

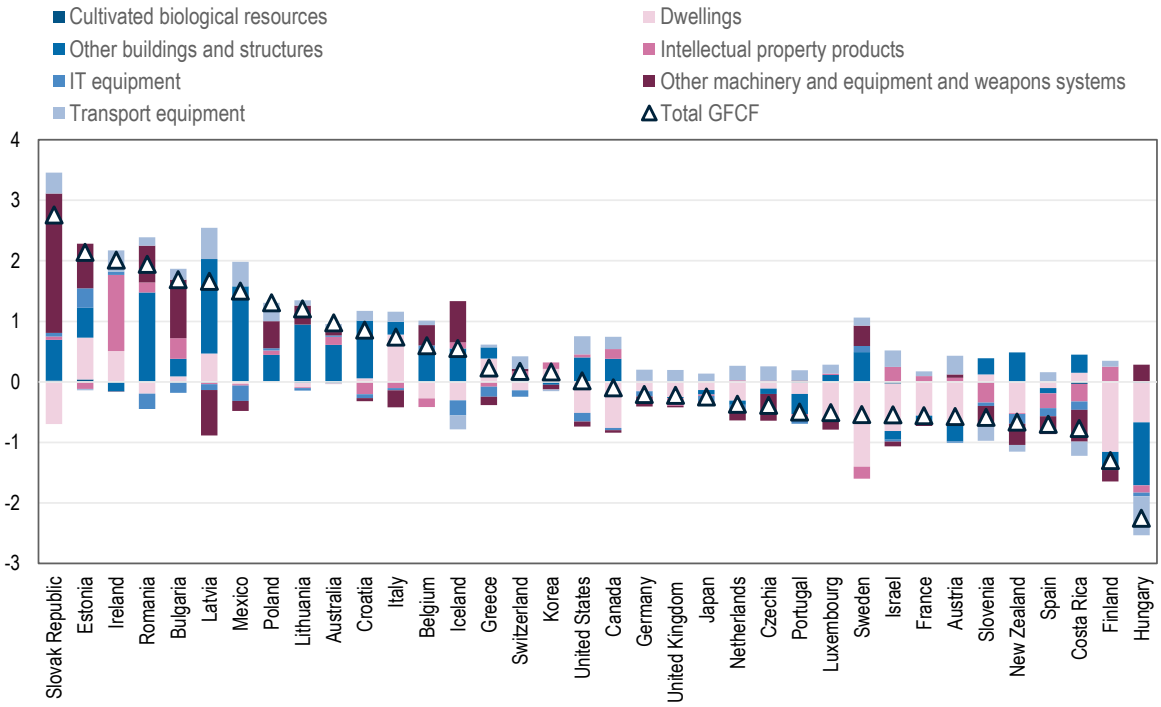


Note: In Australia, ownership transfer costs are included in total GFCF but not allocated to specific assets, causing the sum of GFCF by asset to be lower than the total. In Korea, ICT equipment is included in Other machinery equipment and weapons systems. In Canada and the United States, total GFCF excludes GFCF in Cultivated biological resources.
Source: OECD Productivity Database (2025).

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Figure 3.3. Annual change in investment rate by asset type

2023, Total economy, Percentage points

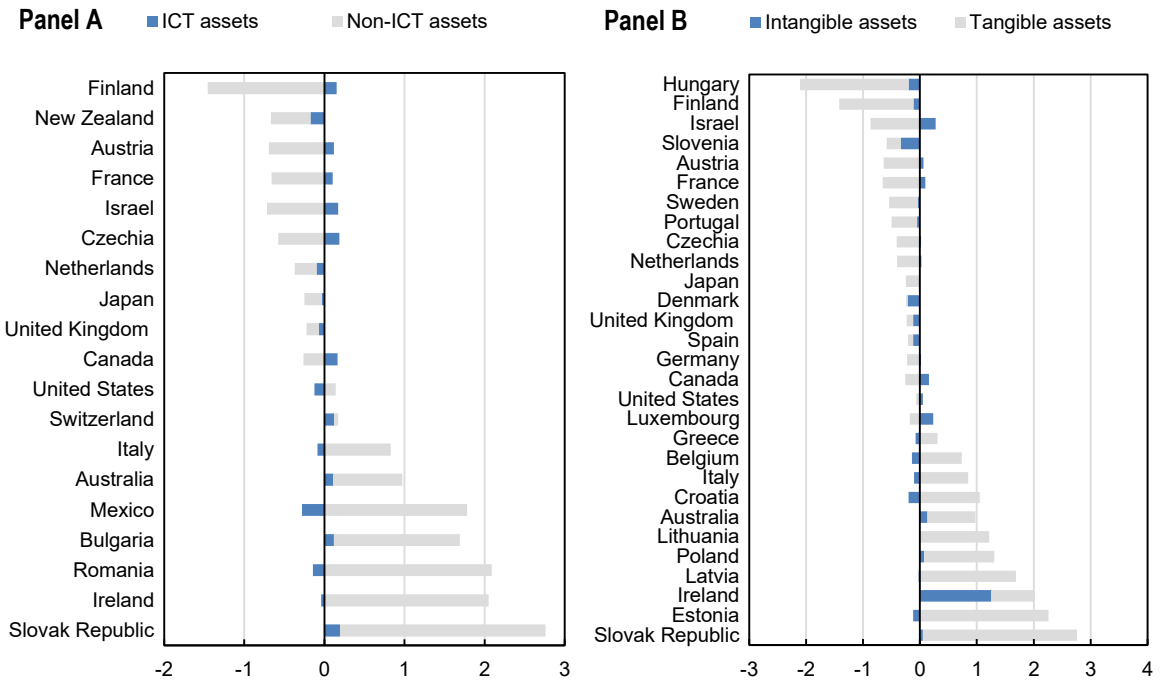


Source: OECD Productivity Database (2025).

StatLink https://stat.link/v4tok6

Figure 3.4. Difference in investment rate by asset type

2023, Percentage points



Source: OECD Productivity Database (2025).

StatLink https://stat.link/umqbw5

How to read the indicators?

The 2008 System of National Accounts (SNA, 2009[4]) recommended a minimum level of asset breakdown (Table 3.1). For the measurement of capital stocks and capital services, homogeneous asset groups with similar price deflators and depreciation patterns are required. The 2008 SNA enlarged the asset boundary by capitalising expenditures in weapons systems and research and development (R&D), which were previously considered as intermediate consumption.

The asset breakdown differs from the one recommended by the 2008 SNA for a few countries. In Korea, ICT equipment is included in Other machinery equipment and weapons systems. In Australia, ownership transfer costs are included in total GFCF but are not allocated across assets. Consequently, the sum of GFCF for individual assets is lower than total GFCF for this country. In Norway, total GFCF excludes investment in weapons systems. In Colombia and Türkiye, the classification of GFCF by type of asset is not available. Therefore, only total GFCF is presented. In Canada and the United States, total GFCF excludes GFCF in Cultivated biological resources.

As discussed in Chapter 5, the 2025 update of the System of National Accounts (SNA, 2025[1]) expands the asset boundary to include additional intangible assets, most notably data, cloud services, Artificial Intelligence, and digital platforms as subcomponents of intellectual property products. These changes aim at improving the relevance of capital measurement in the context of a digital economy. For further information, consult the description of the OECD Productivity Database (OECD, 2025[5]) and the OECD Productivity Statistics – Methodological notes at: www.oecd.org/sdd/productivity-stats/OECD-Productivity-Statistics-Methodological-note.pdf.

Table 3.1. Breakdown of fixed capital assets according to the System of National Accounts 2008

Tangible assets	**N111**	**Dwellings**	**Non-ICT assets**
	N112	**Other buildings and structures**	
	N11M	**Machinery and equipment and weapons systems**	
	N1131	Transport equipment	
	N1132	ICT equipment	**ICT assets**
	N11321	Computer hardware	
	N11322	Telecommunications equipment	
	N110	Other machinery and equipment and weapons systems	**Non-ICT assets**
	N115	**Cultivated biological resources**	
Intangible assets	**N117**	**Intellectual property products**	
	N1171	Research and development (R&D)	
	N1172	Mineral exploration and evaluation	
	N1173	Computer software and databases	**ICT assets**
	N1174	Entertainment, artistic and literary originals	**Non-ICT assets**
	N1179	Other intellectual property products	

Source: OECD elaboration based on the 2008 System of National Accounts.

Data sources

OECD National Accounts Statistics (database), http://data-explorer.oecd.org/s/1v7.

OECD Productivity Statistics (database), http://data-explorer.oecd.org/s/1v8.

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4. Cross-country comparisons of labour productivity levels

Context

This chapter focuses on labour productivity, which is often used to assess convergence across countries. Labour productivity, as used in this chapter, refers to GDP divided by hours worked, or by employment when data on hours are unavailable.

Size of output

Gross Domestic Product (GDP) is a widely used measure of output in the compilation of productivity indicators. It measures the value added generated by an economy, i.e. the value of goods and services produced during a given period, minus the value of intermediate consumption used in the production process. Countries measure GDP in their own currencies. To compare these estimates across countries, they have to be converted into a common currency. The conversion is often made using nominal exchange rates, but these can provide a misleading comparison of the true volume of goods and services produced across countries. A better approach is to use Purchasing Power Parities (PPPs), which are currency converters that control for differences in price levels between countries and allow for international comparisons of the volume of GDP and of the size of economies (European Union/OECD, 2024[1]).

Key findings

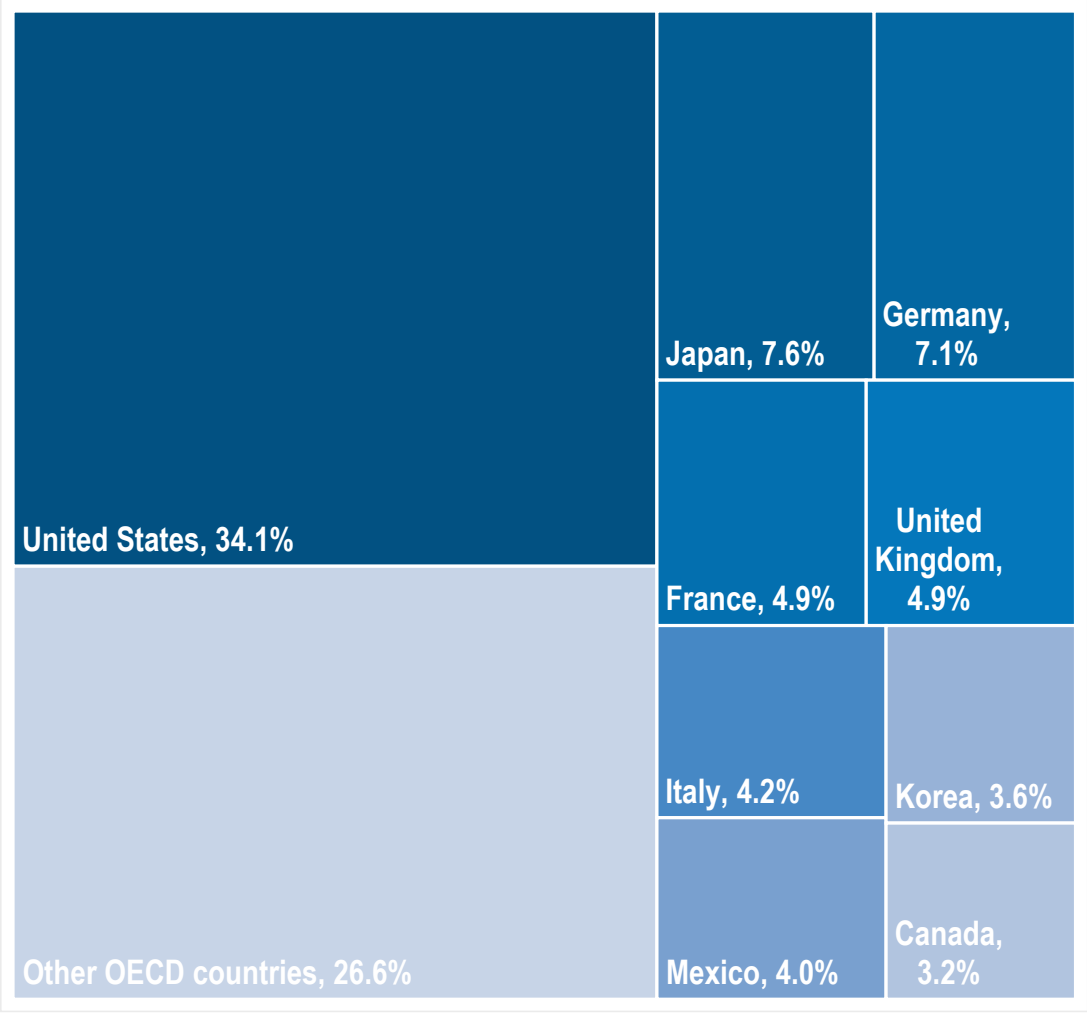
- **Using Purchasing Power Parities (PPPs) instead of nominal exchange rates to convert GDP into US dollars (USD) narrows the gap between most OECD countries and the United States**. In 2023, PPP conversion more than doubled the USD value of GDP in Colombia, Hungary, and Poland, and nearly tripled it in Türkiye. In contrast, Iceland and Switzerland saw slightly lower GDP values in USD when using PPPs, with differences of less than 5% compared to nominal exchange rates.
- **When using PPPs to convert countries' GDP into a common currency, the United States represented the largest share of total OECD GDP in 2023, accounting for over one-third of all OECD countries' combined output**. It was followed by Japan, Germany, France, the United Kingdom, Korea, Mexico, and Canada. The remaining 19 OECD countries together contributed nearly one-quarter of the total OECD GDP.

- **When using nominal exchange rates, the ranking of countries and their shares in the total OECD GDP remain broadly consistent with those obtained using PPPs**. The United States, Japan, and Germany consistently hold the top three positions regardless of the conversion method. However, there are some shifts in the rankings: Germany and Japan swap second and third places, while the United Kingdom and France switched between fourth and fifth, depending on whether PPPs or nominal exchange rates are used.

Indicators

Figure 4.1. Relative size of OECD economies based on GDP current PPPs

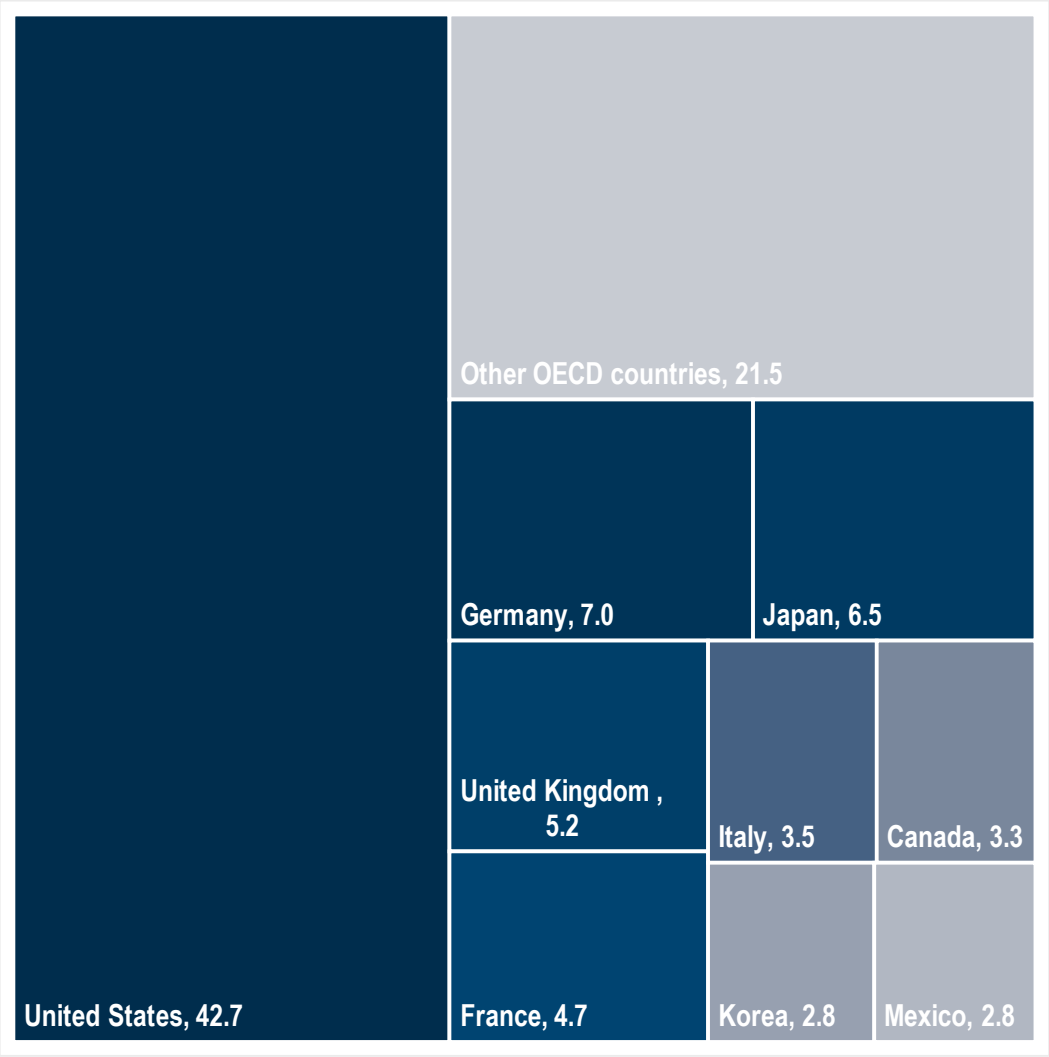
% of OECD total, 2023



Source: OECD Productivity Database (2025).

Figure 4.2. Relative size of OECD economies based on GDP current exchange rates

% of OECD total, 2023



Source: OECD Productivity Database (2025).

How to read the indicators

The compilation of GDP is based on harmonised accounting concepts and definitions that ensure its comparability across countries. In practice, however, the measurement of GDP can be affected by three main issues:

- *The measurement of the non-observed economy*. An exhaustive coverage of production activities can be difficult to achieve in some countries and national estimates may differ in their coverage of non-observed activities. The size of the non-observed economy is generally larger in emerging-market and developing economies reflecting, in part, the higher degree of informal activities and employment.
- *International production arrangements*. In recent decades, globalisation has led to a fragmentation of production processes across countries. In some cases, national accounts record output in the

country where intellectual property products (IPP) are located rather than in the country where output is physically produced (e.g. in the case of contract manufacturing). This can create a disconnection between GDP and production factors, as well as to changes in GDP due to the relocation of IPP from one country to another. Moreover, some of the income generated by IPP may be ultimately transferred abroad - for instance, when affiliates of multinational enterprises hold these assets on their balance sheets and transfer the related earnings to their parent companies (UNECE, 2015[2]). These arrangements can contribute to a growing divergence between GDP and Gross National Income (GNI), as GNI accounts for cross-border property income flows. The 2025 System of National Accounts explicitly highlights how these IPP relocations can change macroeconomic indicators such as GDP and GNI by disconnecting economic ownership from physical production (SNA, 2025[3]).

- *The measurement of the digital economy.* The digital transformation also poses many challenges to the measurement of the production of goods and services and hence GDP. The emergence of new digital services, the increasing scale of peer-to-peer interactions through digital intermediary platforms, the development of "free" services blurring the distinction between consumers and producers, are only a few examples of the challenges currently faced by national accountants (Ahmad and Schreyer, 2016[4]; Ahmad, Ribarsky and Reinsdorf, 2017[5]; UNECE, 2023[6]). Moreover, shorter life cycles of ICT products exacerbate long standing challenges regarding the distinction between price movements and quality increases (Aeberhardt et al., 2020[7]).

Looking forward, harmonisation in these three areas is expected to be enhanced through the implementation of the 2025 System of National Accounts.

With respect to GDP measurement in volume or real terms (i.e. excluding the impact of inflation), the System of National Accounts recommends the production of estimates based on annually chain-linked volume indices. Most countries covered in the report derive annual estimates of real GDP using annually chain-linked volume indices (i.e. updating every year the prices used to measure volume indices). The United States and Canada use chain-linked Fisher indices while other OECD countries use the chain-linked Laspeyres ones. However, Mexico and South Africa currently produce fixed-base volume indices (i.e. measuring volume indices at the prices of a fixed given period) with the base year updated less frequently.

For further methodological information, consult the *OECD Productivity Statistics – Methodological notes* at www.oecd.org/sdd/productivity-stats/OECD-Productivity-Statistics-Methodological-note.pdf and the update of the sources and methodologies used to construct the new OECD Productivity Database (OECD, 2025[8]).

Hours worked and employment

In productivity analysis, the volume of labour input is most appropriately measured by the total number of hours actually worked, i.e. hours effectively used in production, whether paid or not.

In theory, it is preferable to use total hours worked as a measure of labour input in productivity analysis, as variations in working time patterns (e.g. part-time or full-time employment) and employment legislation (e.g. statutory working hours) across countries and over time affect the comparability of total employment figures. However, total employment (i.e. the number of persons employed) is often used as a proxy for labour input, particularly when data on total hours worked are not available.

The relevant concept for measuring labour input is hours actually worked, as opposed to hours paid, contractual hours, or usual hours worked. Hours actually worked reflect regular hours worked by full-time and part-time workers, paid and unpaid overtime, hours worked in additional jobs, excluding time not

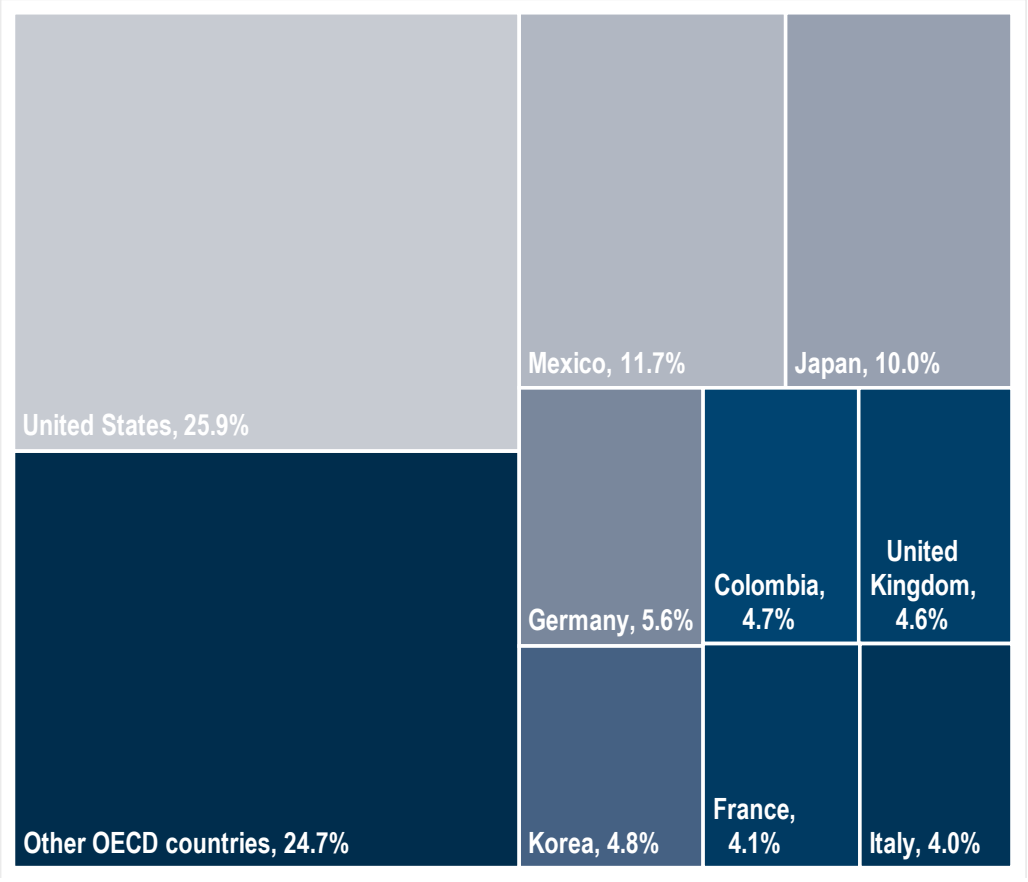
worked for reasons such as public holidays, annual paid leave, sick leave, maternity leave, strikes, bad weather, and economic conditions.

Key findings

- **The United States accounted for one quarter of total hours worked and total employment in 2023, both the largest shares in the OECD area**. However, the ranking of countries in terms of their share in total labour input varies depending on the measure of labour input used, i.e. hours worked or employment.
- **Estimates of average hours worked per worker differ substantially across countries.** While some countries recorded more than 2 000 hours worked per worker in 2023 (such as Colombia, Costa Rica and Mexico), others recorded less than 1 500 hours (Germany, Denmark, Norway, Austria, Sweden, Netherlands, Iceland, Luxembourg, France and Finland). In Germany and Denmark, this figure is below 1 400 hours.
- **Differences in average hours worked per worker across countries partly reflect structural differences in the organisation of labour markets and differences in the method used to measure hours** (Ward, Zinni and Marianna, 2018[9]) (see How to read the indicators for further details).

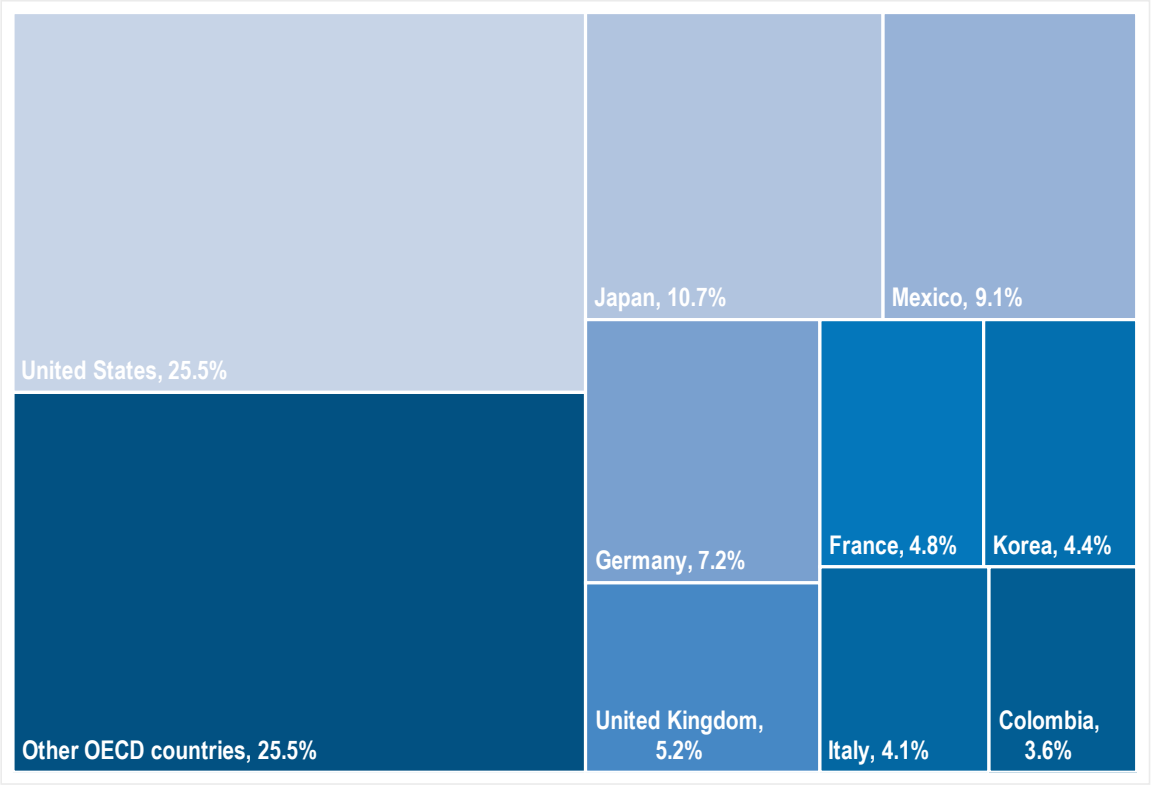
Indicators

Figure 4.3. Relative size of the workforce in OECD economies, based on hours worked, 2023



Source: OECD Productivity Database (2025).

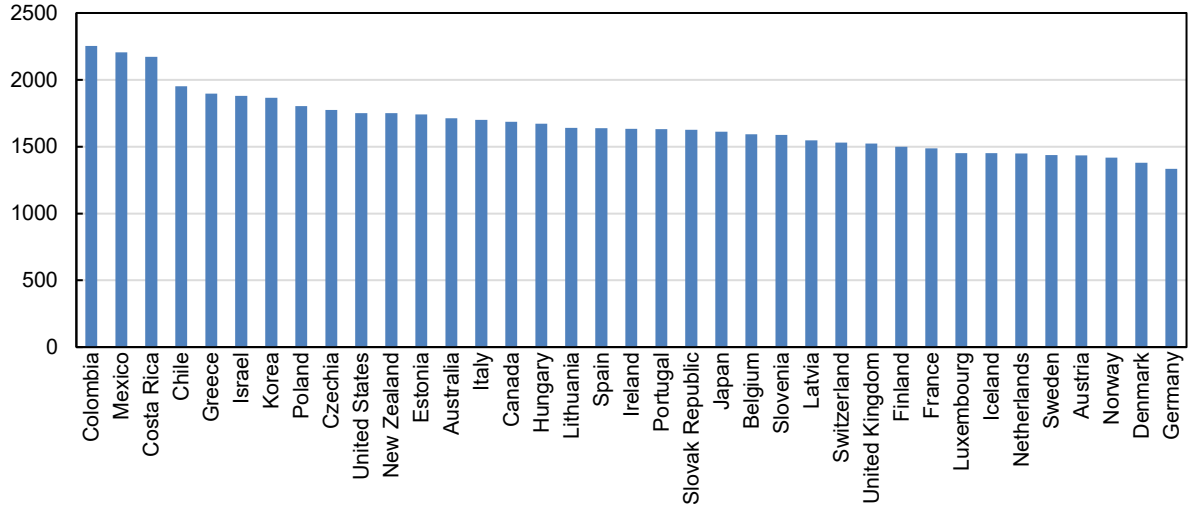
Figure 4.4. Relative size of the workforce in OECD economies, based on employment, 2023



Source: OECD Productivity Database (2025).

Figure 4.5. Average hours worked per worker across countries, 2023

Average annual hours worked per worker



Source: OECD Productivity Database (2025).

StatLink https://stat.link/mhl08d

How to read the indicators

In most countries, the main source to construct measures of hours actually worked is the labour force survey. However, many countries rely on establishment surveys and administrative sources as either primary or auxiliary sources. While the use of different sources may affect the comparability of labour productivity levels, comparisons of labour productivity growth are less likely to be affected (Ward, Zinni and Marianna, 2018[9]).

Computing estimates of hours worked also implies adjusting the activities covered by the labour input measures (employment and hours worked) to those covered by the output measure. This requires adapting the geographical and economic boundaries of employment and hours worked to the national accounts production boundary, in order to exclude resident persons working in non-resident production units and include non-resident persons working in resident production units.

In practice, countries adopt one of two methods to estimate actual hours worked for productivity analysis:

- The **direct method**, which takes actual hours worked self-reported by respondents in surveys, generally labour force surveys (LFS).
- The **component method**, which starts from contractual, paid or usual hours per week from establishment surveys, administrative sources or the LFS, with subsequent adjustments for absences and overtime, and other adjustments to align hours worked with the concepts of hours actually worked and the concept of domestic output.

While the direct approach appeals due to its simplicity, it depends heavily on respondent recall, cannot account for response bias, and assumes a perfect alignment of measures of workers and output. The component approach is more complex, but it systematically attempts to address these issues. There is evidence that response bias and insufficient adjustments to align with the concept of domestic output lead to systematic upward biases in estimates of average hours worked per worker based on the direct approach, as compared to the component approach (Ward, Zinni and Marianna, 2018[9]).

Admittedly, the OECD simplified component method necessarily relies on available data sources. In particular, it assumes that workers in all countries take, on average, all the leave to which they are entitled. However, actual take-up leave rates are likely to reflect differences in working cultures across countries. For this and other reasons, like the access of national statistics offices to a variety of national data sources, the OECD simplified component method estimates can be considered only as a stopgap for those countries currently using a direct approach with minimal or no adjustments, while these countries work towards improving their methodologies (Ward, Zinni and Marianna, 2018[9]).

Finally, the effective quantity of labour input depends not only on the total number of hours actually worked but also on the education, working experience, business functions and other workers' characteristics. The measure of labour input used in this publication, i.e. total hours worked, does not account for the composition or "quality" of the workforce and likely underestimates the effective contribution of labour to production.

For further methodological information, consult the *OECD Productivity Statistics – Methodological notes* at www.oecd.org/sdd/productivity-stats/OECD-Productivity-Statistics-Methodological-note.pdf and the update of the sources and methodologies used to construct the new OECD Productivity Database (OECD, 2025[8]).

Labour productivity

Labour productivity is the most frequently computed productivity indicator. It represents the volume of output produced per unit of labour input. The ratio between output and labour input depends to a large extent on the presence of other inputs, such as physical capital (e.g. buildings, machinery and transport vehicles) and intangible assets used in production (e.g. intellectual property products), technical efficiency and organisational change.

Intangible assets play an important role in economic growth and productivity. This raises measurement challenges related to the potential recording of output where intellectual property products are located, rather than where output is physically produced. In addition, IPP may give rise to large income transfers between the countries where they are registered, often for fiscal reasons, such as low-tax jurisdictions, and those where the ultimate owners reside. This can result in a significant gap between GDP and GNI (Gross National Income). In such cases, measures of labour productivity based on GNI are more meaningful than those based on GDP.

The 2025 System of National Accounts (SNA, 2025[3]) gives more prominence to intangible assets through dedicated chapters on digitalisation and globalisation. The former enhances the visibility of digital activities in macroeconomic accounts and expands the scope of intangible assets by classifying data as productive assets within a newly established category named “data and databases”, and providing statistical definitions of cloud services, Artificial Intelligence, and digital platforms. The latter clarifies the classification, valuation, and ownership of intellectual property products within multinational enterprises, including cross-border flows of output and income.

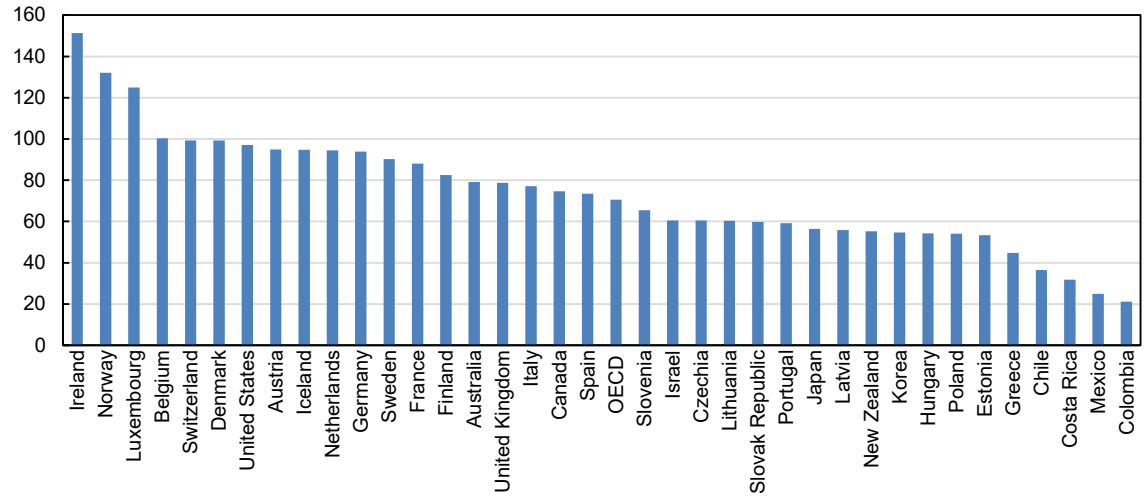
Key findings

- **There are large disparities in labour productivity levels across OECD countries.** Measured as GDP per hour worked in PPP terms, average labour productivity across OECD countries was around USD 70 per hour in 2023. The highest-performing economies recorded labour productivity levels nearly twice the OECD average, while those at the lower end were around one third of the average (Figure 4.6). While patterns are broadly similar, the difference is less marked when using GNI per hour worked, which better reflects the role of multinational enterprises (MNEs) (Figure 4.8). In terms of disparities, the maximum-to-minimum labour productivity ratio reveals that the most productive countries in the OECD required seven times less labour to produce the same output as the least productive one on average.
- **Labour productivity levels across OECD countries have continued to converge since 2000, especially due to gains in countries that initially had the lowest productivity levels**. Since 2000, most countries with labour productivity levels below the OECD average have narrowed the gap (Figure 4.7). However, by 2023, Israel, Japan, and Greece experienced further declines in relative productivity, moving even further below the OECD average - showing a diverging pattern. Among countries that initially had above-average productivity, most have moved closer to the OECD average, with the exception of Ireland and Iceland, where productivity rose even further above the average.
- **In most countries, labour productivity measures based on GDP and GNI are similar, as cross-border income flows are relatively small or tend to offset each other.** However, in countries such as Ireland and Luxembourg, significant differences emerge due to income generated abroad, particularly by multinational enterprises. In these cases, GNI-based measures provide a more accurate reflection of national productivity. Finally, labour productivity measures based on Gross Value Added (GVA) yield lower values than those based on GDP, as they exclude product taxes, and subsidies. Overall, GVA-based measures generally follow a distribution that mirrors GDP-based metrics (Figure 4.8).

Indicators

Figure 4.6. Labour productivity in 2023

GDP per hour worked in current prices and current PPPs

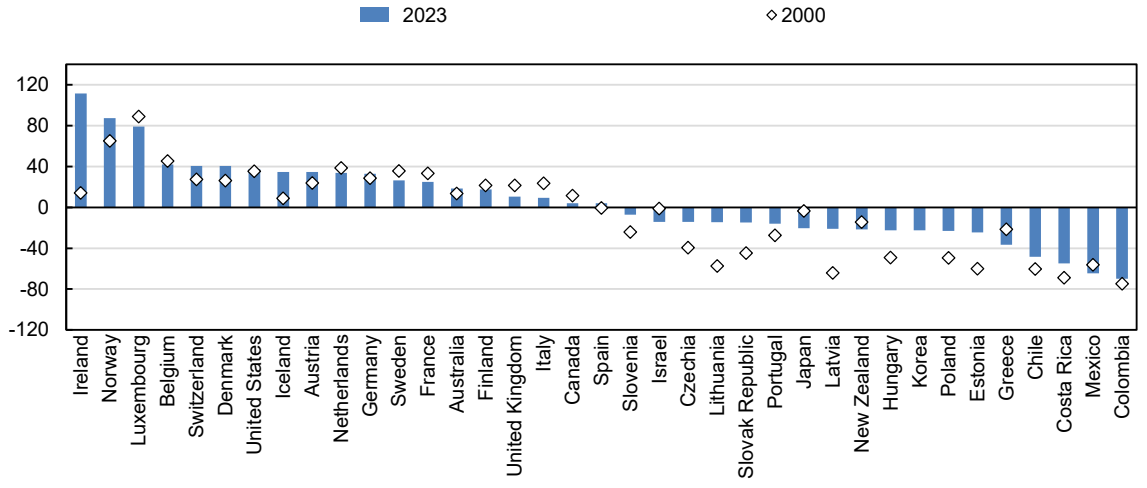


Source: OECD Productivity Database (2025).

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Figure 4.7. Labour productivity gap relative to the OECD average

GDP per hour worked in current prices and current PPPs

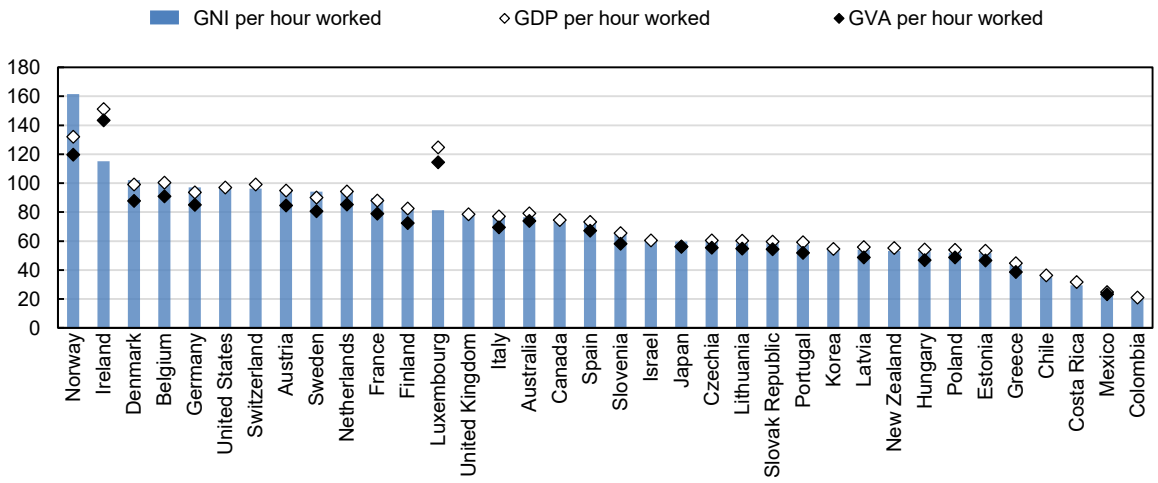


Source: OECD Productivity Database (2025).

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Figure 4.8. Labour productivity levels based in 2023.

GDP, GNI and GVA per hour worked in current prices and current PPPs



Note: GNI per hour worked data for Norway refer to 2022.
Source: OECD Productivity Database (2025).

StatLink https://stat.link/cyu6sp

How to read the indicators

Following national accounts standards, and consistently with the measure of output, the measure of labour input in an economy includes the contribution of cross-border workers working in resident production units. Conversely, it excludes all persons working in non-resident production units. Depending on the original data sources used to estimate employment (e.g. labour force survey, administrative data, business statistics), various adjustments are needed to ensure consistency between labour and output measures.

In Figures 4.6 to 4.8, national accounts data on hours worked for Austria, Estonia, Finland, Greece, Latvia, Lithuania, Poland, Portugal, Sweden and the United Kingdom have been replaced with estimates obtained with the OECD simplified component method described in the section on Hours worked. However, the impact of this correction on labour productivity growth rates is marginal (Ward, Zinni and Marianna, 2018[9]).

The difference between GDP and GNI depends on where headquarters are located. If an affiliate is established in an investment hub but headquarters remain abroad, the profits shifted to that affiliate are imputed back to the country of the parent company, and GNI remains unaffected. Conversely, if headquarters are set up in the investment hub where profits are artificially inflated, no such imputation takes place. In that case, GNI remains close to GDP unless the profits are actually transferred abroad through dividend payments, which would then reduce GNI (Deaton and Schreyer, 2021[10]).

For further methodological information, consult the *OECD Productivity Statistics – Methodological notes* at www.oecd.org/sdd/productivity-stats/OECD-Productivity-Statistics-Methodological-note.pdf and the update of the sources and methodologies used to construct the new OECD Productivity Database (OECD, 2025[8]).

Data sources

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Ward, A., M. Zinni and P. Marianna (2018), "International productivity gaps: Are labour input measures comparable?", *OECD Statistics Working Papers*, No. 2018/12, OECD Publishing, Paris, https://doi.org/10.1787/5b43c728-en. [9]

5. Industry contributions to aggregate labour productivity growth

Context and key findings

This chapter examines industry contributions to total labour productivity growth with a focus on post-COVID-19 trends.

The analysis covers the total economy, excluding real estate, public administration and defence, education, and health activities. Labour productivity developments at the industry level are broken down into within-industry effects and cross-industry reallocation effects:

- The within-industry effect is affected by underlying developments and characteristics at the industry and firm level. Examples include the intensity with which industries use skilled labour and capital, their capacity to innovate, and their exposure to competition and knowledge sharing through their participation in international trade and global value chains. It can also result from resource reallocation across firms.
- The cross-industry reallocation effect reflects reallocation of hours worked between industries: it is the sum of reallocations between industries with different productivity levels (static reallocation effect) and between industries with different productivity growth rates (dynamic reallocation effect). The static effect usually dominates the dynamic effect.

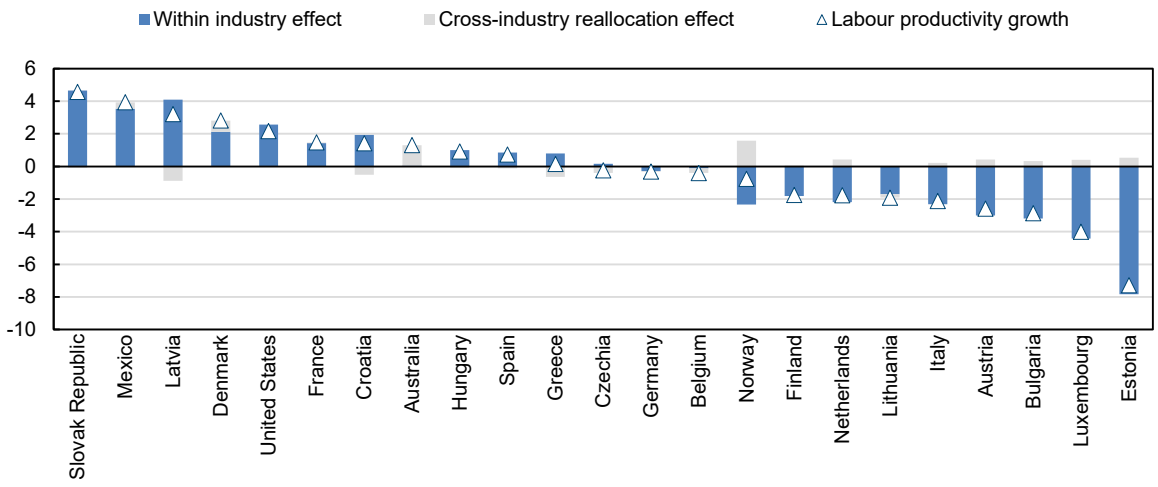
Key findings

- **In 2023 aggregate labour productivity growth was primarily driven by the within-industry effect, which captures overall changes in labour productivity in each industry** (Figure 5.1). This outcome holds for all 23 OECD countries with available data, with the exception of Australia, Belgium, and Czechia. The within-industry effect boosted productivity growth in about half of the countries, while in the other half it dragged the overall labour productivity growth into negative territory. The sharpest drops in within-industry contribution were recorded in Luxembourg and Estonia, while the most significant increases were observed in Slovak Republic, Latvia and Mexico.
- **The impact of cross-industry reallocations on overall labour productivity growth, captured by the changes of hours worked across industries, were marginal in most OECD countries in 2023.** It was smaller than 0.5 percentage points for all except seven countries with data. This pattern is expected as reallocation of hours worked between industries is usually modest during recessions.
- **The overall impact of cross-industry reallocations on labour productivity growth was negative in about half of the OECD countries with available decomposition data in 2023.**

Norway, Australia, Denmark and Estonia experienced the most significant positive effects, while Greece, Latvia and Croatia saw the largest negative contributions. Czechia and Belgium stand out as exceptions, where the cross-industry reallocation effect dominated the within-industry effect. However, the magnitude of this reallocation effect in those countries was small (below 0.4 percentage points).

Figure 5.1. Decomposition of labour productivity growth

2023, Percentage points



Note: The figure shows labour productivity for the total economy, derived by aggregating data across industries. Labour productivity at the industry level can be decomposed into two main components: the within-industry effect and the cross-industry reallocation effect. Within-industry effect refers to productivity improvements within each industry. The cross-industry reallocation effect reflects reallocation of hours worked between industries.
Source: OECD Productivity Database (2025).

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There is no clear pattern in reallocations of hours worked across industries and countries

The limited overall reallocation of hours worked across industries during normal economic circumstances is often the result of industry contributions with opposite signs (Figure 5.2). A positive reallocation effect happens when resources are moving from industries with lower productivity (or its growth rate) towards more productive ones. Conversely, a negative reallocation effect occurs when industries performing below the productivity average attract a relatively larger share of resources.

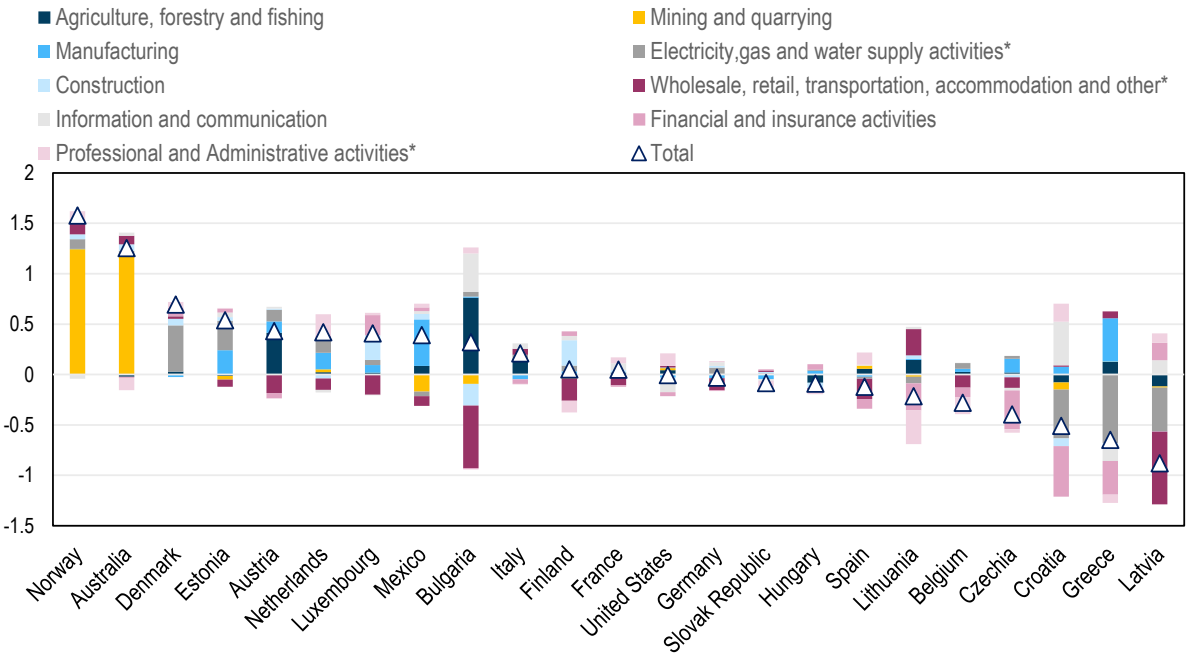
- **In half of OECD countries for which the data is available in 2023, the cross-industry reallocation effect was negative.** However, this aggregate figure masks variation in industry contributions across countries.
- **Lower-productivity sectors influenced the cross-industry reallocation effect in different ways in different economies.** For example, the shift of hours worked from agriculture (Bulgaria and Austria) or construction (Finland) toward more productive industries boosted labour productivity growth. On the contrary, trade and personal services such as wholesale and retail trade, transportation, accommodation and food continued to increase their share of hours worked relative to other sectors, thus exerting a negative influence on labour productivity growth, in more than half of the countries with data. Such a trend has been visible in most countries since 2021.

However, the pace of change has slowed, with the increase in 2023 smaller than in 2022 in many countries. One contributing factor may be the impact of inflation, which dampened consumption of goods and services in these sectors. Services prices, in particular, remained higher than goods prices. This could potentially reflect the lower sensitivity of service prices to commodity price movements and the higher labour intensity of services industries, where wage adjustments – an important driver of service costs – tend to lag behind shifts in economic activity (OECD, 2024[1]). Although household consumption expenditure in EU countries continued to grow in 2023 after adjusting for inflation, the growth rate was markedly slower compared to the previous year (Eurostat, 2023[2]).

- Reallocation effects triggered by high-productivity industries showed some degree of similarity across countries. In 2023, the contribution of digital-intensive industries to aggregate productivity growth lost momentum as already seen in 2022, with the reallocation of hours worked towards the information and communication industry becoming marginal in many cases, and finance activities even showing a strong contraction in hours worked in several countries. Bulgaria and Croatia were exceptions, in that the contribution of the information and communication sector remained significantly positive, similar to 2022.
- Mining and quarrying, typically a high productivity industry due to its high capital intensity and dominance by large global firms, made a strong contribution to the cross-industry reallocation effect and to labour productivity growth in Norway and Australia. Energy-related activities, made up by electricity, gas, steam and air conditioning supply, also drove up labour productivity growth in a number of countries such as Denmark and Estonia, where hours worked in this sector expanded. On the contrary, in countries such as Greece and Croatia, a contraction in the share of hours worked in energy-related activities led to a negative impact on productivity growth.

Figure 5.2. Industry contributions to cross-industry reallocation effect

2023, Percentage points



Note: (*) denotes the aggregated industry. Wholesale, retail, transportation, accommodation and other is the combination of ISIC Rev.4 industries G (Wholesale and retail trade), H (transportation and storage), I (accommodation and food service activities), R (Arts, entertainment and recreation), and S (Other services activities). Electricity, gas and water supply activities is the combination of ISIC Rev.4 industries D (Electricity, gas, steam and air conditioning supply) and E (Water supply; sewerage; waste management and remediation activities).
Source: OECD Productivity database (2025).

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Within-industry reallocations are the main driver of productivity growth

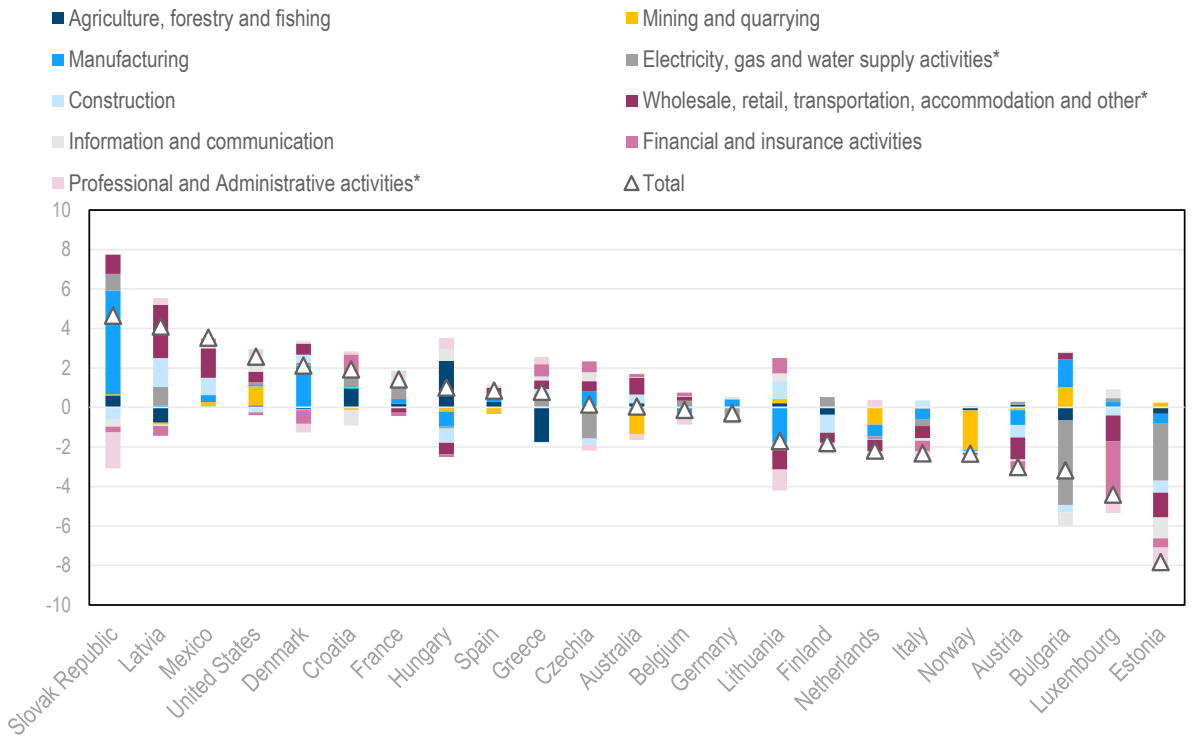
- **Within-industry effects continued to be the key factor underlying the developments in aggregate labour productivity growth in most OECD countries in 2023, as in 2022** (Figure 5.3).
- **The within-industry contribution to productivity growth was negative in 2023 in around half of the OECD countries for which data is available.** In most of those countries, the within-industry contributions of most individual industries were either negative or close to zero. This downward pressure was only slightly offset by small positive contributions from a few industries. In Estonia, for example, productivity declined in all industries except mining and quarrying, which recorded a modest gain in 2023. Whereas Estonia and Luxembourg saw the sharpest within-industry productivity declines, Slovak Republic and Latvia recorded the largest increases.
- **The relative contributions of industries to within-industry development varied across countries, reflecting differences in their industrial structure and business environment.** In around half of the OECD countries with available 2023 data, technology-intensive sectors such as information and communication, as well as finance, continued to exert a drag on within-industry productivity. The emergence of generative Artificial Intelligence (AI) technologies, such as

ChatGPT and Gemini, in late 2022, is expected to have the greatest impact on these sectors given their higher AI exposure and use (OECD, 2024[3]). While the adoption of AI is widely anticipated to boost productivity, the 2023 data suggest that its benefits have yet to materialise (Chapter 1). This delay is not surprising, as the effective integration of AI requires skilled talent, the development of appropriate applications, and time for adoption and diffusion (Filippucci, Gal and Schief, 2024[4]).

- **Trade services, including wholesale, retail, transportation, accommodation and restaurants and other personal services, continued recording positive labour productivity growth in 2023,** contributing significantly to overall productivity gains in several countries. This was particularly evident in Latvia, Mexico and Slovak Republic where within-industry effects in these industries were among the strongest. In contrast, Luxembourg and Estonia saw negative and substantial contributions of these industries.
- The continued growth of labour productivity in trade services may reflect the ongoing digitalisation that has accelerated since 2000. Innovations such as contactless mobile payments, QR-code menus, service robots, and expanded food delivery options have increasingly reshaped these industries (Esposito et al., 2022[5]; Goolsbee et al., 2025[6]). A notable digitalisation trend was also observed in Latvia and Mexico in 2023, potentially explaining the strong labour productivity growth in trade services recorded in both countries that year. In Latvia, digital investment among SMEs grew significantly in 2023, with the share of enterprises possessing at least basic digital skills rising by 12.5% (European Commission, 2024[7]).
- The primary sector, which encompasses agriculture and the mining and quarrying industry, displayed mixed within-industry effects across OECD countries in 2023. While agricultural activities exerted positive within-industry effects in more than half of the OECD countries, the impact on aggregate productivity was small due to their limited weight in the total economy in many countries. The strongest increase in the within-industry effect in agriculture was observed in Hungary, whereas Greece saw the sharpest decline.
- By contrast, mining and quarrying contributed negatively to the within-industry productivity effect in most countries in 2023. This typically occurs when the growth in output volume does not keep pace with the increase in total hours worked, or when output declines faster than working hours are reduced. For example, Australia and Norway – two major mining and quarrying exporters – saw an increase in hours worked in these activities. One possible explanation for these countries' expansion in mining activities in this year is the response to rising global demand for metals and minerals critical to the green energy transition (IEA, 2023[8]). Another contributing factor could be country-level efforts to address global energy shortages, partly driven by ongoing sanctions against Russia. Although investment in mining and quarrying rose in these countries in 2023, output has not yet grown at the same rate as labour input, suggesting a time lag before the benefits of higher investment are fully realised (Productivity Commission, 2024[9]).

Figure 5.3. Industry contributions to within-industry effect, 2023

2023, Percentage points



Note: (*) denotes the aggregated industry. Wholesale, retail, transportation, accommodation and other is the combination of ISIC Rev.4 industries G (Wholesale and retail trade), H (transportation and storage), I (accommodation and food service activities), R (Arts, entertainment and recreation), and S (Other services activities). Electricity, gas and water supply activities is the combination of ISIC Rev.4 industries D (Electricity, gas, steam and air conditioning supply) and E (Water supply; sewerage; waste management and remediation activities).
Source: OECD Productivity database (2025).

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How to read the indicators

The decomposition of aggregate labour productivity growth that is used in this chapter includes three main terms, each of them corresponding to a sum of industry contributions:

- A *within-industry effect*, where labour productivity growth in each industry is weighted by the industry share in total value added in year t-1.
- A *static reallocation effect*, accounting for changes between t-1 and t in the share of total hours worked of industries with different productivity levels. Industries with an increasing share in total hours worked contribute positively to aggregate labour productivity growth if they have an above-average labour productivity level.
- A *dynamic reallocation effect*, accounting for changes between t-1 and t in the share of total hours worked of industries with different productivity growth rates. An increase in the total hours worked share of industries with positive productivity growth has a positive effect on aggregate labour productivity growth. This effect is more significant when the industry value added is high.

For additional information on this decomposition of aggregate labour productivity growth, see the methodological note.

This chapter focuses on a subset of the total economy that excludes real estate, public administration and defence, education, and health activities (i.e. total economy less industries L, O, P, Q in the ISIC rev. 4 classification). Real-estate activities are excluded because their value added is largely imputed (it includes the value of both actual and imputed housing rents in the economy) and disproportionate as compared to the corresponding work force in national accounts (mostly real-estate agents and employees of notary offices are attached to the real-estate industry in national accounts). Public administration and defence, education and health services are excluded because they are largely non-market. Hence, their output value is measured as the sum of input costs, and in several countries their output volume is measured by deflating input costs, thus conventionally excluding any productivity gains.

For most countries, the above decompositions of aggregate labour productivity growth rely on breakdowns of value added and hours worked into industries at the two-digit level of the NAICS 2017 classification (for Canada, Mexico, and the United States) or the ISIC rev. 4 classification. Due to data limitations, the decompositions for France, Germany, Italy, and the United Kingdom rely on a mix of one-digit and two-digit level data corresponding to the A38 level of the ISIC rev.4 classification, and the decomposition for Australia relies on one-digit level data according to the ANZSIC 2006 classification. This corresponds to between 20 and 64 industries, depending on data availability in different countries. Except for a few cases, this is the most granular industry data publicly available in national accounts, but it might not be sufficient to fully capture the heterogeneity of economic activities. Therefore, it cannot be excluded that part of the within-industry effects presented above actually correspond to resource reallocations between firms or economic activities belonging to the same two-digit industry. A complete assessment of the contribution of reallocations and business dynamism (entries and exits of firms) to aggregate labour productivity growth would require firm-level data.

Even though a more granular breakdown is used for all calculations, the following breakdown by industry is used to visualise the contributions of reallocation and within-industry effects to aggregate labour productivity growth in the figures included in this Chapter:

- agriculture and mining: industries A and B in the ISIC rev. 4 classification; industries 11 and 21 in the NAICS 2017 classification
- manufacturing and utilities: industries C, D and E in the ISIC rev. 4 classification; industries 22 and 31-33 in the NAICS 2017 classification
- construction: industry F in the ISIC rev. 4 classification; industries 23 in the NAICS 2017 classification
- trade services: industry G, H, I and R to U in the ISIC rev. 4 classification; industries 42,44RT, 48TW, 71, 72 and 81 in the NAICS 2017 classification
- finance and insurance: industry K in the ISIC rev. 4 classification; industry 52 in the NAICS 2017 classification
- business services: industries M and N in the ISIC rev. 4 classification; industries 54 to 56 in the NAICS 2017 classification.

Data sources

OECD National Accounts Statistics (database), http://data-explorer.oecd.org/s/1v7.

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6. Industry-level multifactor productivity growth

Context and key findings

Multifactor productivity (MFP) is an estimate of the overall efficiency with which inputs are used in the production process. It is calculated as a residual, using a growth accounting framework developed by Solow (1957[1]). This framework enables the decomposition of changes in economic output – in this case gross value added (GVA) – into contributions from changes in labour and capital inputs, and a residual known as MFP. More information on its derivation can be found in OECD (2025[2]).

Being a residual, changes in MFP reflect a mix of factors such as technological change, changes in capacity utilisation or economies of scale, or measurement errors. MFP is often associated with technological progress. Innovations, such as the use of Information and Communication Technology in production, alongside complementary assets like organisational and human capital, enables a more efficient production process (Bajgar, Criscuolo and Timmis, 2021[3]; André and Gal, 2024[4]). For instance, the adoption of computers and the expansion of the internet are considered as a key driver of MFP growth in OECD countries during the period from the mid-1990s to the early 2000s (Kretschmer, 2012[5]; Cette, Devillard and Spiezia, 2020[6]).

MFP estimates may also capture measurement errors and are particularly sensitive to how input and output volumes are measured. For instance, inaccuracies in measuring input volumes can arise when differences in input quality are not accounted for, or when various types of investment assets are not properly distinguished (Pionnier, Zinni and Baret, 2023[7]). Similarly, measuring output volumes presents challenges – especially when related to households, the non-market sector, and the informal economy – where gauging output values is often difficult (OECD/APO, 2022[8]).

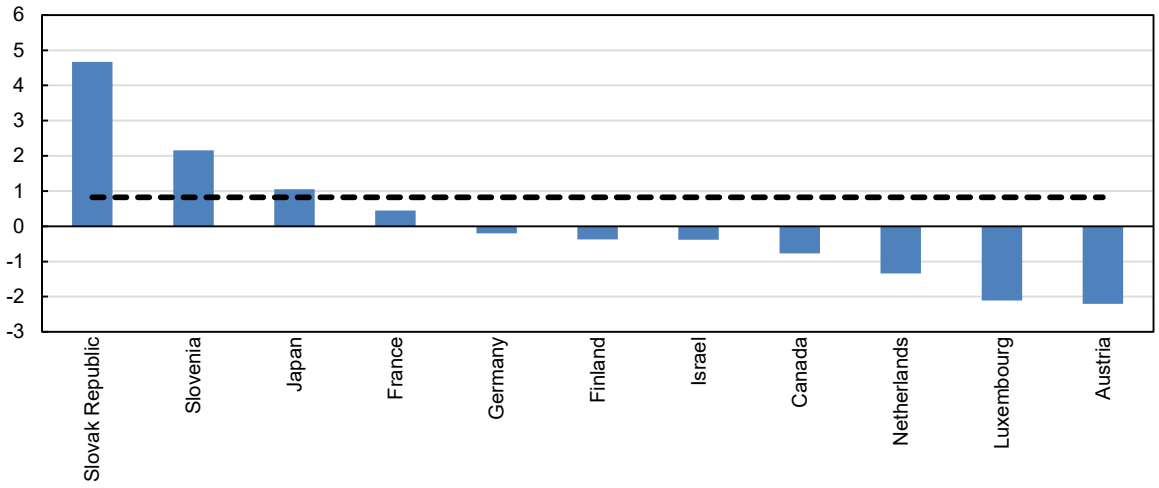
MFP growth can also reflect cyclical fluctuations in the economy, and it has often been found to be pro-cyclical, i.e. declining during economic downturns and rising during expansions. This pattern is partly due to unmeasured variations in the utilisation of labour and capital over the business cycle (OECD, 2015[9]). Since adjusting inputs can be costly, firms often respond to short-term changes in demand by altering the intensity of use of existing resources – for instance, by hoarding labour during downturns or underutilising capital without immediately reducing capacity. Hence, adjusting MFP estimates to account for these cyclical effects enhances their usefulness by isolating the structural component of productivity growth, including technological progress (Fernald, 2014[10]; Martin and Jones, 2023[11]; Comin et al., 2025[12]) and Chapter 8 on cyclical adjustment of MFP).

Key findings

- **Total economy MFP growth either stagnated or even turned negative in most countries with available data for 2023** (Figure 6.1). The largest declines were recorded in Austria and Luxembourg, where MFP growth fell by around -2%. In contrast, the highest MFP growth in 2023 was observed in the Slovak Republic and Slovenia, reaching approximately 4.5% and 2%, respectively. Most countries with available data in 2023 recorded a lower MFP growth than the long-run average MFP growth across OECD countries, with the exception of the Slovak Republic, Slovenia, and Japan. Against the backdrop of the introduction of ChatGPT and Gemini in early 2022, the subdued MFP performance suggests that any productivity gains from generative AI have yet to materialise (Filippucci et al., 2024[13]).
- However, **the economy-wide average masks significant variation across industries.** Mining and quarrying and financial services recorded negative MFP growth in most countries in 2023. The decline of MFP in these industries was mainly driven by a large fall in GVA coupled with limited adjustment in inputs - namely hours worked and capital services. Similar patterns were reported in Australia and Canada for these industries (Australian Bureau of Statistics, 2023[14]; Statistics Canada, 2025[15]). Manufacturing showed little to no MFP growth, except in the Slovak Republic and Luxembourg, which saw a notable increase. In contrast, the energy and information and communication industries experienced increases in MFP in more than half of the countries with data. Nevertheless, their positive contribution remained modest due to the relatively small weight of these industries in the total economy in most countries (Figure 6.2).
- Preliminary industry-level data for 2023 indicate that **in most business economy industries, MFP growth was lower than prior to the COVID crisis** (2010-2019). Notable exceptions were the information and communication industry, where MFP growth continued to outpace pre-COVID levels in more than half of the countries with data (Figure 6.3). The deceleration in MFP growth relative to the pre-COVID period was evident as early as 2022 across several business economy industries, with the trend intensifying in 2023. In mining and quarrying as well as manufacturing, the post-COVID rebound in 2021 proved short-lived, as MFP gains decelerated steadily over the following two consecutive years. Similarly, activities such as finance and professional activities experienced a loss of momentum, with MFP growth in 2023 falling below the relatively stronger performance seen in 2022. The only notable exception is the information and communication industry, where MFP growth in 2023 remained above pre-pandemic levels in half of the countries.
- **MFP growth explained the overwhelming part of labour productivity growth in 2023 in most countries and industries.** The dominance of MFP contribution over capital deepening has been evident in the past for long (OECD/APO, 2022[8]). For the G7 average, the sharpest declines in MFP growth were observed in mining and trade services (including wholesale and retail trade, transportation, and accommodation and food services). By contrast, the information and communication industry registered a positive MFP contribution to labour productivity growth of around 1.8 percentage points.
- **The contribution of capital deepening, while still being negative, showed a slight improvement compared to its level in 2022** (Figure 6.4). The stagnation of capital deepening growth in 2023, in particular within the G7, was partly driven by the slowdown of capital services. Capital services, which reflect accumulated investment net of depreciation from wear, tear, and obsolescence, have stagnated in part due to weak investment trends in these countries (Chapter 4). In addition, their rate of depreciation has been increasing, in part because of the rising share of asset classes with shorter service lives, such as ICT and intangible assets (Sicsic, 2018[16]).

Figure 6.1. MFP growth for the total economy

2023, Per cent

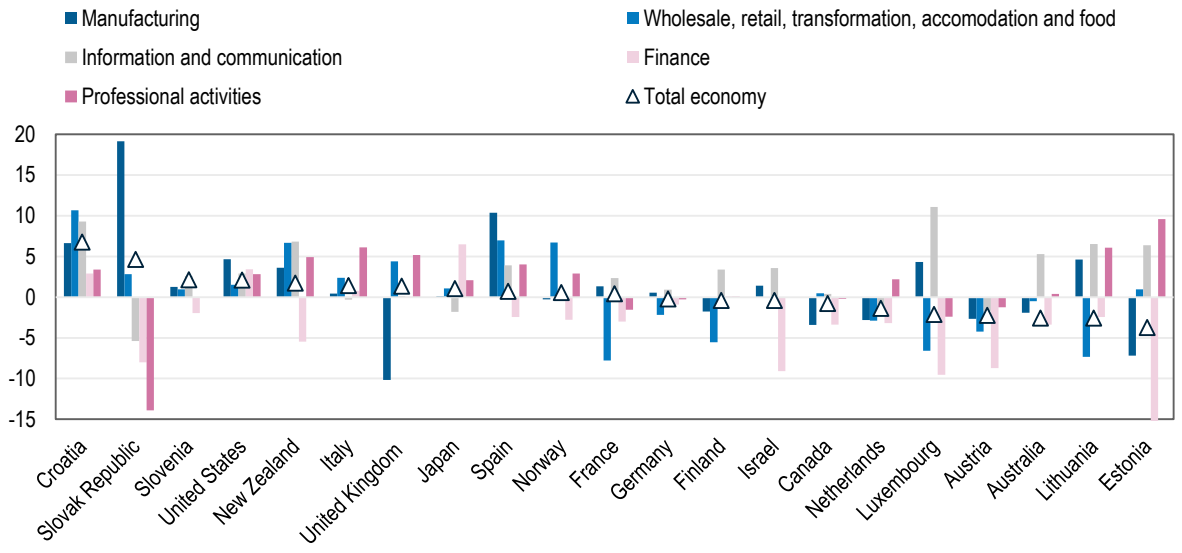


Note: MFP estimates are produced for countries that provide detailed industry-level data on investment, capital stock, and hours worked, and that participated in the one-off OECD data collection on asset depreciation and retirement profiles. The dotted line represents the average MFP growth across OECD countries with available data over 2010-19.
Source: OECD Productivity database (2025).

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Figure 6.2. MFP growth across industries compared to the total economy

2023 or latest available year, Per cent

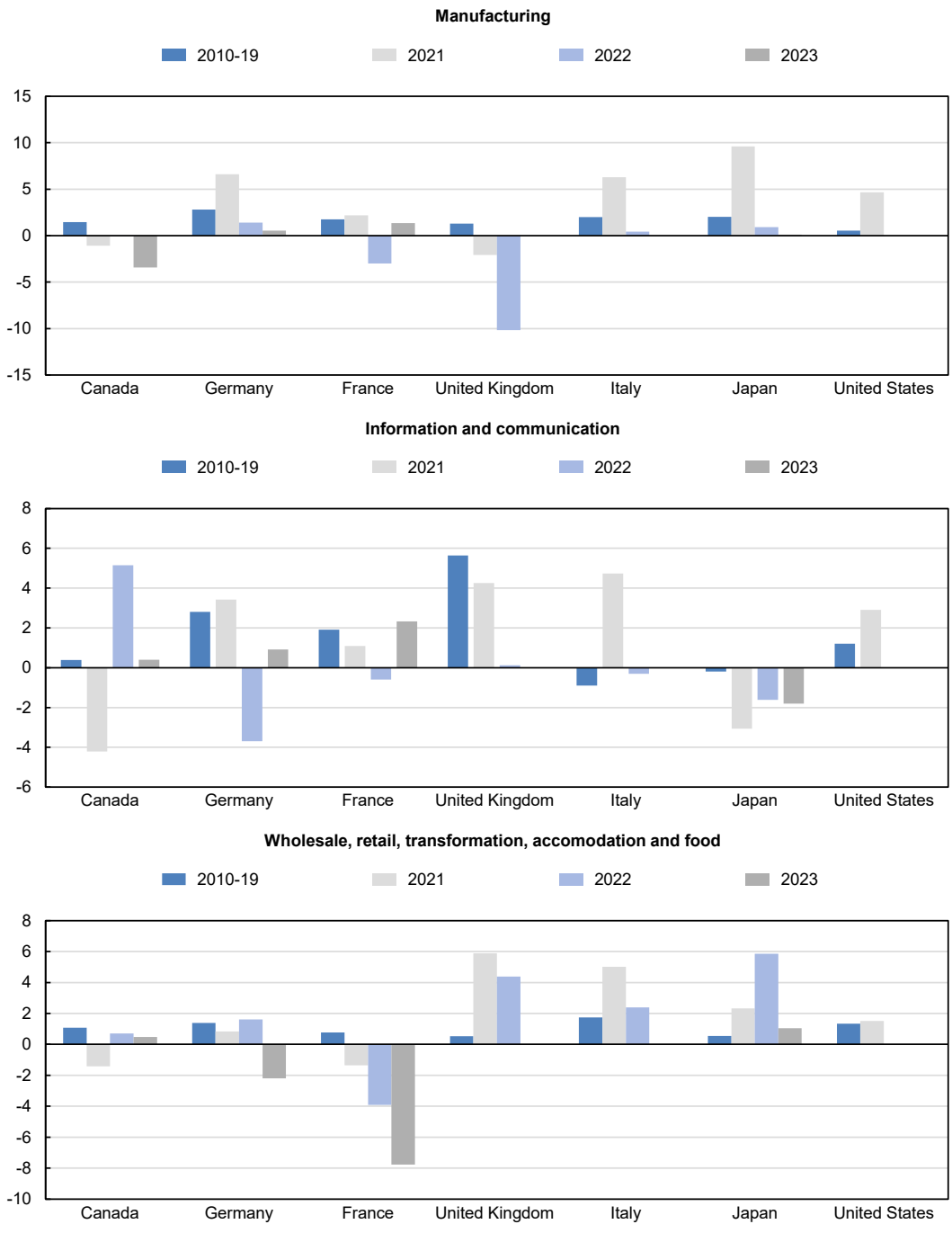


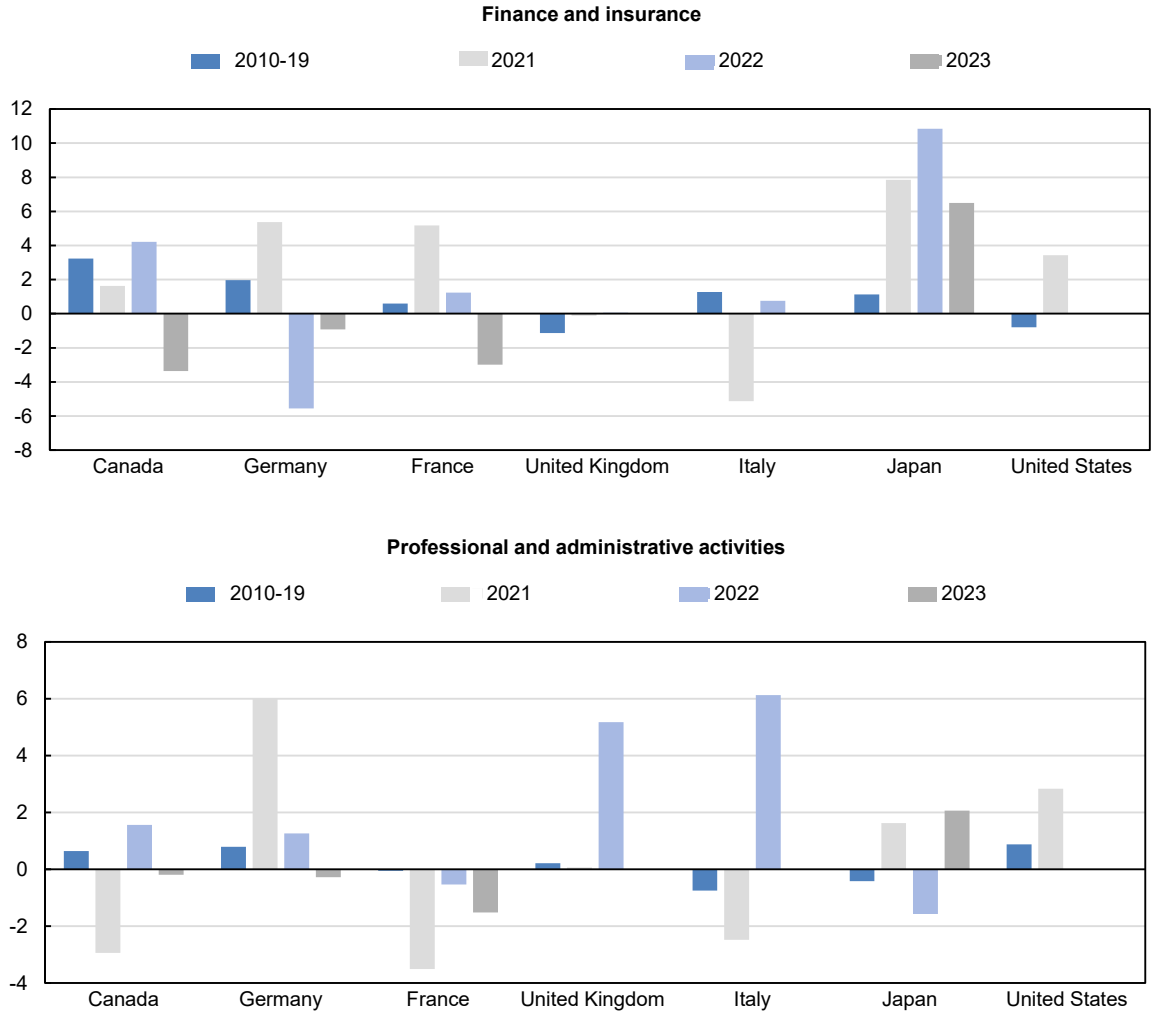
Note: Industry-level MFP estimates are produced for countries that provide detailed industry-level data on investment, capital stock, and hours worked, and that participated in the one-off OECD data collection on asset depreciation and retirement profiles. Countries for which data refers to 2022: Australia, Estonia, the United Kingdom, Croatia, Italy, Lithuania, Norway. Countries for which data refers to 2021: Spain, New Zealand and the United States.
Source: OECD Productivity database (2025).

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Figure 6.3. MFP growth in selected industries across G7 countries

Per cent



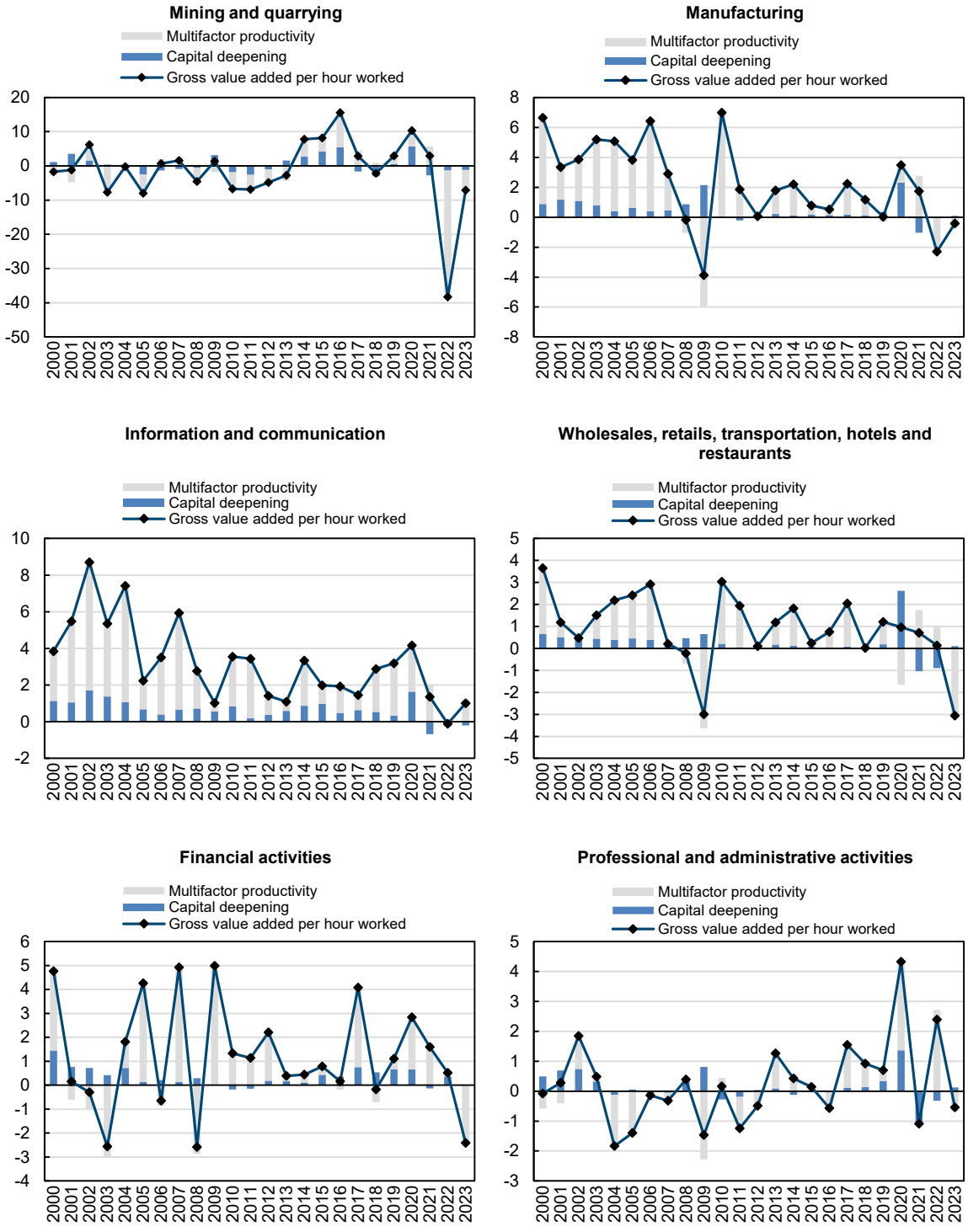


Note: 2023 figures do not include data for United Kingdom, Italy, and United States, whose data have not been released at the time of writing.
Source: OECD Productivity database (2025).

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Figure 6.4. MFP contribution to labour productivity in selected industries: G7 average

Percentage points



Note: 2023 figures do not contain data for United Kingdom, Italy, and United States, as data for these countries have not yet been released.
Source: OECD Productivity database (2025).

StatLink https://stat.link/3il9k7

Data sources

OECD Productivity Database: http://data-explorer.oecd.org/s/1v8.

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7. Productivity in SMEs and large firms

Context and key findings

A large body of evidence documents differences in productivity across firms (Cette, Corde and Lecat, 2018[1]) and (Cunningham et al., 2022[2]). In several countries, a fat tail of low-productivity firms - composed in large part of small firms - coexists with large firms that are highly productive and exposed to international competition. To the extent that large firms can exploit increasing returns to scale, productivity tends to increase with firm size. Moreover, large firms tend to adopt new technologies more readily than small firms, although this is not necessarily the case for new or younger firms. Large firms also have easier access to finance and to foreign markets. The industry in which a firm operates is a further major determinant of its productivity performance.

While new small firms can also spur aggregate productivity growth when they exploit new technologies and stimulate productivity-enhancing changes by incumbents, severe economic downturns may lead to a missing generation of start-ups (OECD, 2023[3]). While this effect is usually marginal in the short term, the absence of these start-ups may affect long-term productivity, as they play a key role in competition, innovation (Kolev et al., 2022[4]), as well as in the diffusion of new technologies and business models (Criscuolo, Gal and Freund, 2024[5]).

Scale-up dynamics significantly influence firm productivity. Firms that scale up through employment tend to become more productive during their high-growth phase, gradually catching up with peers. In contrast, firms that scale via turnover often expand their workforce just before growth, which temporarily reduces productivity, but their productivity improves as employment stabilises during the growth phase (OECD, 2021[6]). Human capital (e.g. workforce skills, management skills) is another key factor that explains differences in productivity across firms (Criscuolo, Gal and Freund, 2024[5]; Criscuolo et al., 2021[7]).

In addition, the presence of highly productive multinational enterprises (MNEs) can also impact national productivity figures. On the one hand, they may create opportunities for local firms to benefit from knowledge spillovers (Javorcik, 2004[8]; Alfaro-Ureña, Manelici and Vasquez, 2022[9]). On the other hand, profit-shifting by MNEs to low-tax jurisdictions can artificially inflate productivity figures for such countries and complicate cross-country comparisons (Bricongne, Delpeuch and Lopez-Forero, 2023[10]).

Key findings

- **Large firms generally exhibit higher labour productivity than smaller companies in most OECD countries.** Luxembourg stands out as an exception, with very small firms with 1-9 employees surpassing the largest firms in labour productivity. The largest gaps in labour productivity levels in 2023 or for the latest available year – measured as value added per person employed – between large and smaller firms were observed in Ireland and Norway (Figure 7.1). Such a gap in Ireland is mainly due to an over-representation of very large and highly-productive firms – mostly foreign-owned – which benefit from low statutory income tax rates (OECD, 2022[11]). In Norway, the high labour productivity of very large firms was particularly evident in the mining and quarrying sector. This was likely driven by a surge in exploration activities of oil and petroleum over the past few years (OECD, 2024[12]). In Europe, multinationals are also found to be more productive than other firms, with no marked difference between domestic and foreign-owned multinationals (Eurostat, 2025[13]).

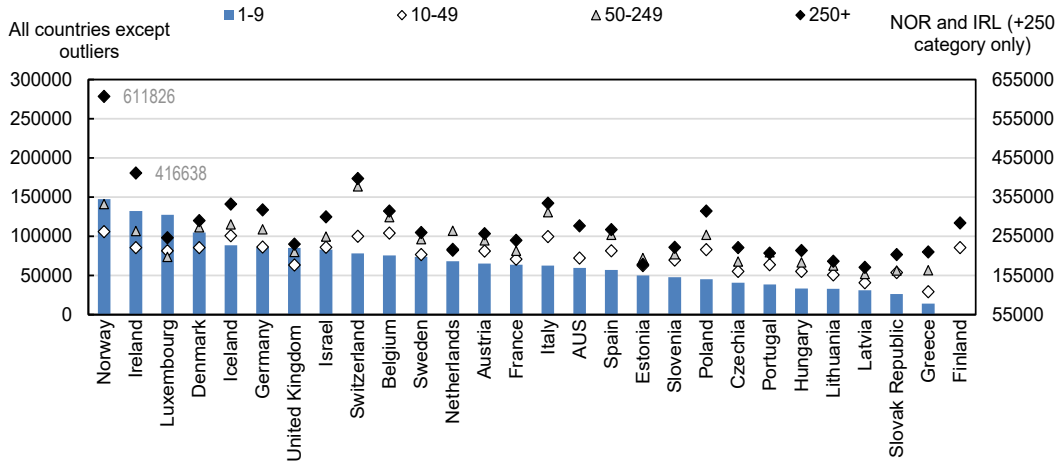
- **On average across countries, firms with more than 250 employees produce about twice as much output per hour as firms with 10-19 employees.** Within each country, large firms also tend to be about 75% more productive than mid-sized firms (50-249 employees), and around one-third more productive than firms with 100-249 employees. These patterns point to substantial productivity differences by firm size, even within national economies.

- **In manufacturing, the productivity gap between large and small firms was on average more pronounced than in the business economy as a whole, reflecting returns to scale from capital-intensive production** (Figure 7.2). This strong size–productivity link in manufacturing is consistent with previous findings (Berlingieri, Calligaris and Criscuolo, 2018[14]). Among OECD countries for which data is available, manufacturing firms with less than ten employees in Türkiye and Greece had the biggest productivity gap relative to outsized enterprises. In Ireland, the productivity gap of medium-sized with respect to large enterprises was similar to that of small firms, and the biggest among OECD countries. Conversely, the productivity of small and medium-sized enterprises in Japan was the closest to that of large firms, even exceeding it in case of medium-sized firms.
- **Compared with manufacturing, differences in labour productivity between firm size classes are less pronounced in business services**. Firm size tends not to be the sole determinant of firm's performance in terms of productivity in business activities (Berlingieri, Calligaris and Criscuolo, 2018[14]). In some OECD countries, smaller firms even outperformed large ones in terms of labour productivity (see Figure 7.3). This is likely attributed to the competitive advantages of small-size firms in these activities compared to their larger counterparts, such as agility, niche focus, specialised expertise, and flexibility (Hallward-Driemeier, Nayyar and Davies, 2022[15]; Marchese et al., 2019[16]). In addition, startups may possess valuable intangible assets – such as R&D or patents – that are highly scalable at limited marginal cost (Haskel and Westlake, 2018[17]). These advantages are likely to enable start-ups and young firms to scale up – meaning increase their turnover – without necessarily sizing up their capital stock or expanding their workforce at the same rate (Audretsch, Kritikos and Schiersch, 2020[18]; Mula, Zybura and Hipp, 2024[19]).

Indicators

Figure 7.1. Labour productivity in SMEs and large firms, business economy

USD per person employed, current prices, 2023 or latest available year

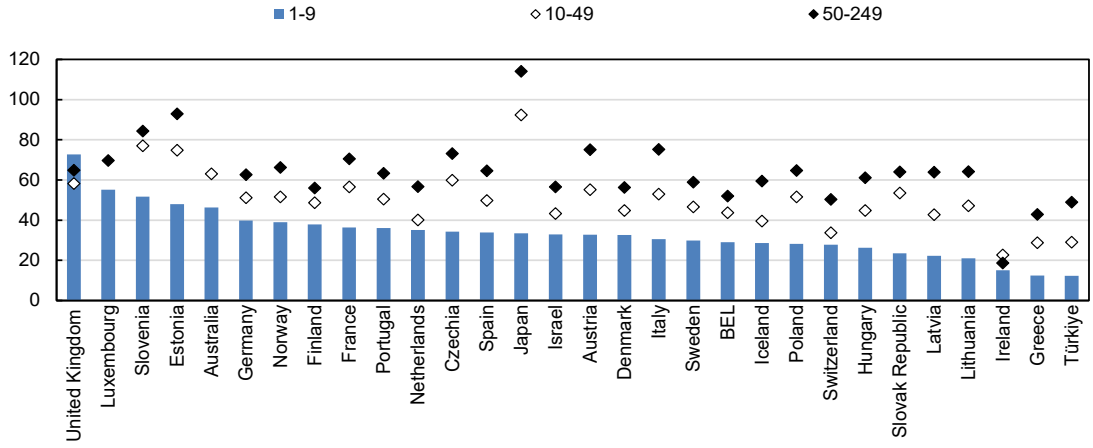


Note: 1-9 corresponds to firms with less than 10 employees, 10-49 corresponds to firms having more than 10 but less than 50 employees, 50-249 corresponds to firms with more than 50 but less than 250 employees, 250+ corresponds to firms with more than 250 employees.
Source: OECD Structural and Demographic Business Statistics (database).

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Figure 7.2. Labour productivity in SMEs and large firms, manufacturing

Index = 100 for firm size 250+, 2023 or latest available year

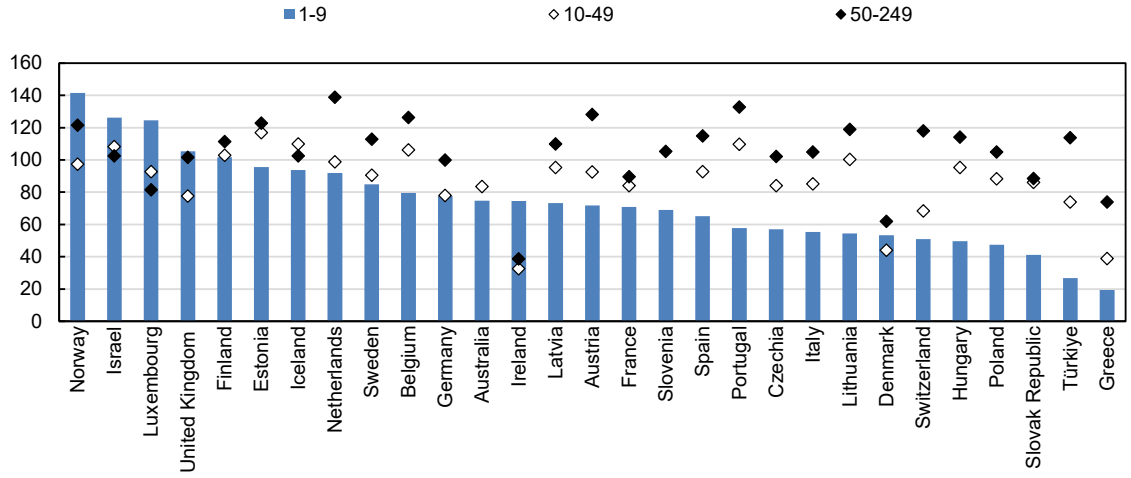


Note: 1-9 corresponds to firms with less than 10 employees, 10-49 corresponds to firms having more than 10 but less than 50 employees, 50-249 corresponds to firms with more than 50 but less than 250 employees.
Source: OECD Structural and Demographic Business Statistics (database).

StatLink https://stat.link/zk15dt

Figure 7.3. Labour productivity in SMEs and large firms, business services

Index for firm size 250+, 2023 or latest available year



Note: 1-9 corresponds to firms with less than 10 employees, 10-49 corresponds to firms having more than 10 but less than 50 employees, 50-249 corresponds to firms with more than 50 but less than 250 employees.
Source: OECD Structural and Demographic Business Statistics (database).

StatLink https://stat.link/eod87l

How to read the indicators

Labour productivity by enterprise size-class is measured as gross value added at current prices per person employed. Labour input is measured as total employment, which includes employees and all other paid or unpaid persons who worked during the reference year. Data on hours worked by all persons employed are typically not available by enterprise size class.

Value added and total employment for different enterprise size classes are sourced from OECD Structural and Demographic Business Statistics (database). They usually do not perfectly align with the corresponding estimates in national accounts. For instance, the latter include adjustments to capture the informal economy. Since labour input is measured as total employment, the cross-country comparability of labour productivity measures by size class may also be affected by differences in the share of part-time employment. In addition, productivity differences in the main aggregate sectors could mask different productivity patterns in more narrowly defined industries.

In this chapter, “business economy” covers mining and quarrying (B), manufacturing (C), electricity, gas, steam and air conditioning supply (D), water supply, sewerage, waste management and remediation activities (E), construction (F) and business services (excluding finance and insurance activities). Business services include wholesale and retail trade, repair of motor vehicles and motorcycles (G); transportation and storage (H); accommodation and food services (I); information and communication services (J); real estate activities (L); and professional, scientific, administrative and support activities (M and N) (letters between brackets correspond to the industry codes in ISIC rev. 4).

Data sources

OECD Structural and Demographic Business Statistics (database), http://dx.doi.org/10.1787/sdbs-data-en.

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8. Cyclical adjustment of multifactor productivity

Context and key findings

A broad-based revival in productivity growth is crucial to addressing long-term socio-economic challenges across OECD countries. Productivity growth – producing more with the same or fewer inputs – is the primary engine of sustained economic growth and rising living standards (André and Gal, 2024[1]). In today's context, where labour force expansion is slowing and capital accumulation is constrained by high debt and uncertainty, input-driven growth is limited. That puts a spotlight on multifactor productivity (MFP) as the key engine of sustainable growth. MFP isolates the portion of productivity growth not explained by increases in labour or capital inputs – it reflects how efficiently these inputs are used (Chapter 2). This includes gains from technological progress, innovation, improved skills deployment, better resource allocation, and organisational or managerial efficiency.

Historically, MFP has demonstrated a strong correlation with overall economic growth, primarily because measured productivity often rises during economic expansions and declines during recessions. However, this procyclicality largely arises from fluctuations in factor utilisation, such as labour hoarding during downturns or intensified capital usage during expansions. Recognising this is crucial to distinguish underlying productivity trends from transient cyclical movements. Failing to account for these cyclical variations risks misinterpreting temporary shocks as structural changes, potentially leading to inaccurate assessments of longer-term productivity developments. Using a range of techniques, this chapter provides insights into the evolution of cyclically-adjusted MFP growth in the manufacturing sector for 20 OECD countries (listed in Annex Table 8.A.1) since the end of the 1990s.

Key findings

- **Experimental estimates suggest that cyclically-adjusted multifactor productivity growth (MFP) averaged around 2.2% across a sample of OECD countries between 1995 and 2022,** close to the average unadjusted MFP over the same period. The findings appear robust to the methodological choices and consistently highlight a trend of weak, stagnating, or continuously declining MFP growth in recent decades.
- **By filtering out short-term fluctuations, cyclical adjustment offers a distinct view of productivity trends that complements the picture from unadjusted data, especially during downturns.** In the 2008 crisis, for example, adjusted MFP estimates (0.1% to 2.6%) across selected OECD countries were notably higher than the unadjusted rate (-2.5%), illustrating how this alternative lens enhances analytical depth.

- **Adjusted MFP trends over the last 20 years show a broad slowdown across the European Union, the United States and Japan.** Growth fell from 2.4-3.7% in the European Union to 0.5-2.2% (with major divergences among countries) compared to 3-5.8% to 0.7-1.3% in the United States and from 2% to -0.8% in Japan.

Cyclically-adjusted MFP growth in the manufacturing industry

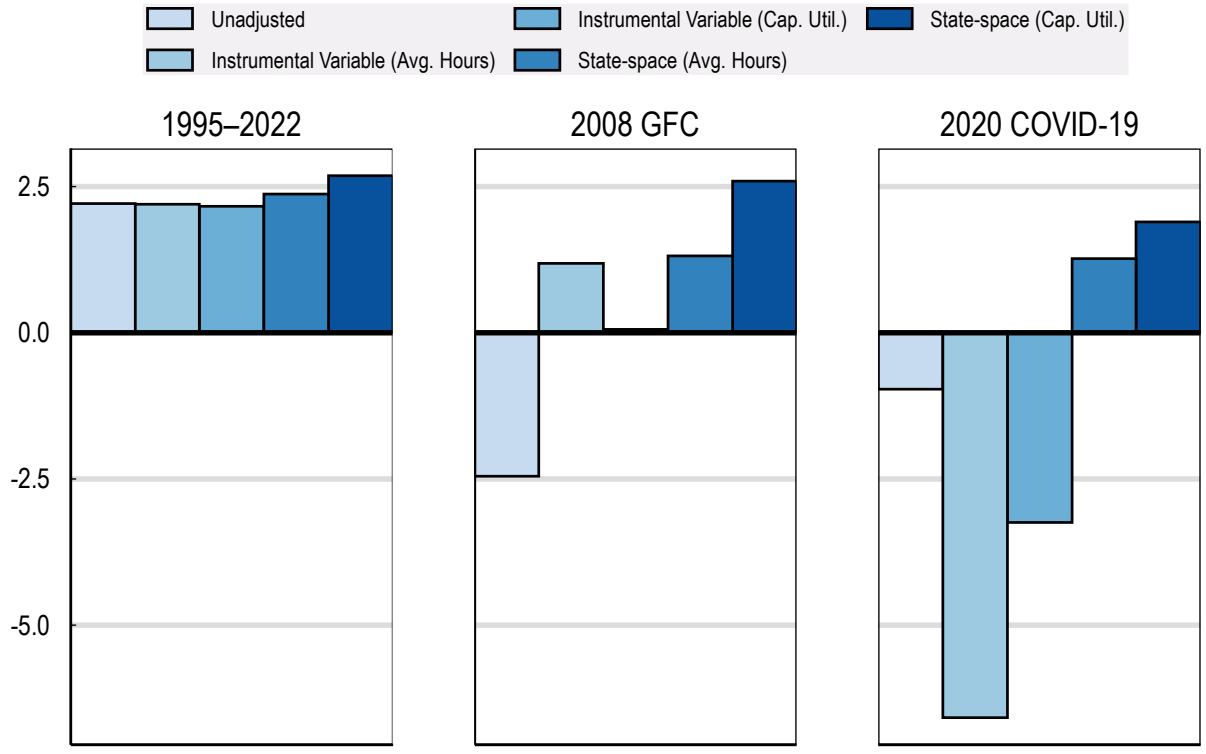
The manufacturing industry plays a crucial role in analysing Multifactor Productivity (MFP) due to its capital intensity and its central role in shaping economic progress. Yet, manufacturing productivity exhibits significant procyclicality – rising sharply during expansions and declining in recessions – due to high fixed costs, substantial economies of scale, and labour hoarding practices (Basu, Fernarld and Kimball, 2006[2]). The high degree of tradability and extensive input-output relationships in manufacturing render this part of the economy especially exposed to the spread of cyclical shocks across industries and countries (Engel and Wang, 2011[3]; Ariu, 2016[4]; Miyamoto and Nguyen, 2024[5]). This underscores the importance of distinguishing between manufacturing productivity movements driven by short-term volatility from those reflecting long-term structural change.

Between 1995 and 2022, average unadjusted MFP growth across selected OECD countries was 2.2%. To better capture the underlying evolution of productivity, two adjustment methodologies, instrumental variable (IV) regression and state-space modelling, were tested, each relying on two distinct proxies for input utilisation: capacity utilisation rates from business tendency surveys and average hours worked per worker. IV regressions show little sensitivity to the choice of proxy, with both yielding an average of adjusted MFP growth of 2.2% across selected OECD countries. State-space models produced marginally higher estimates (2.4%-2.7%) for the same countries and period. Given the traditionally large uncertainties around MFP estimates, the close alignment across methods lends confidence to the robustness of the adjusted MFP results.

Overall, these findings indicate that cyclical adjustment techniques consistently provide reliable estimates of underlying structural MFP trends, regardless of the specific modelling approach or input utilization proxy employed. The benefits of cyclically adjusting MFP are particularly evident when analysing periods of economic recession and expansion. For instance, at the peak of the 2008 financial crisis, average unadjusted MFP growth across selected OECD countries declined sharply to -2.5%. In contrast, cyclically-adjusted estimates remained notably higher: IV regression adjustments yielded estimates of 0.1% (capacity utilisation) and 1.2% (average hours worked), while state space models ranged between 1.3% and 2.6%. Conversely, during the economic expansion preceding the crisis (2002–2007), unadjusted MFP growth for selected OECD countries averaged as high as 4.1%, but cyclically adjusted estimates were notably lower: IV regression adjustments provided estimates of 2.7% (capacity utilisation) and 2.8% (average hours worked), while state-space models indicated slightly higher estimates, ranging between 3.6% and 3.7%.

Figure 8.1. Comparison of cyclically-adjusted MFP growth

Average across selected OECD countries, Percentage points



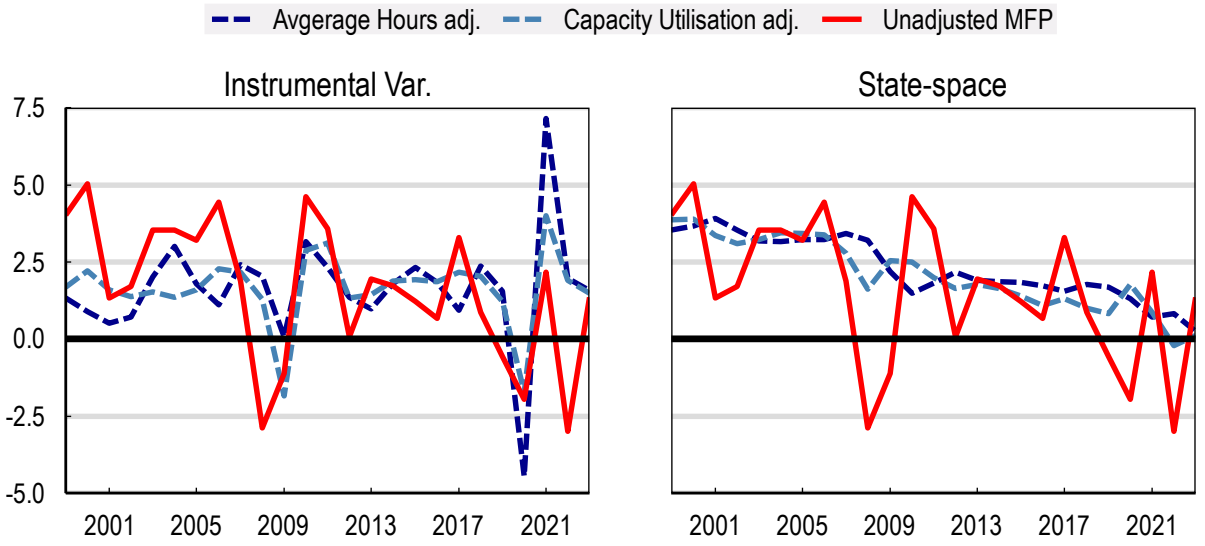
Note: This chart compares average Multifactor Productivity (MFP) growth estimates for 18 OECD countries (Australia, Austria, Germany, Estonia, Finland, France, Spain, the United Kingdom, Italy, Japan, Lithuania, Israel, Luxembourg, the Netherlands, Norway, Slovenia, the United States, and Slovak Republic) using different cyclical adjustment methods. "GFC" refers to the 2008 Global Financial Crisis, while "COVID-19" indicates the pandemic-related economic downturn in 2020. Adjustments are made using two modelling approaches: Instrumental Variable (IV) regression and state-space modelling. Both methods utilise proxies for input utilisation—either capacity utilisation ("Cap. Util.") based on business tendency surveys, or average hours worked per worker ("Avg. Hours").
Source: (Arend, Botev and Jaax, forthcoming[6]).

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During the more recent period of volatility associated with the COVID-19 pandemic, the two adjustment techniques diverged significantly. While average unadjusted MFP growth rates across selected OECD countries dropped to -1%, IV regression adjustments intensified the procyclical effects, producing significantly lower estimates ranging from -3.3% to -6.7%. Conversely, the state space model continued to filter out cyclical volatility effectively, yielding positive estimates between 1.3% and 1.9% (Figure 8.1). Visual analysis of country-specific adjusted series further highlights inherent methodological trade-offs (Figure 8.2). Although IV regression adjustment aligns closely with actual MFP growth series and thus offers more conservative estimates, it struggles during exceptional periods, such as the COVID-19 pandemic. This limitation arises from instruments selected primarily based on long-term cyclical correlations, which might fail to adapt adequately to extraordinary circumstances. Conversely, state space model adjustments, assuming a more pronounced cyclical component, provide greater adjustments during volatile periods, effectively isolating only very long-term productivity trends.

Figure 8.2. Comparison of cyclically adjusted MFP growth for France

MFP growth, Percentage points



Note: The blue lines represent the cyclically adjusted MFP series based on average hours worked per worker, while the dark blue lines represent the cyclically adjusted MFP series based on capacity utilisation rates. The dashed lines indicate the cyclical adjustment using instrumental variables, whereas the dot-dashed lines show the adjustment using a state-space model. The red line represents the unadjusted MFP series.
Source: (Arend, Botev and Jaax, forthcoming[6]).

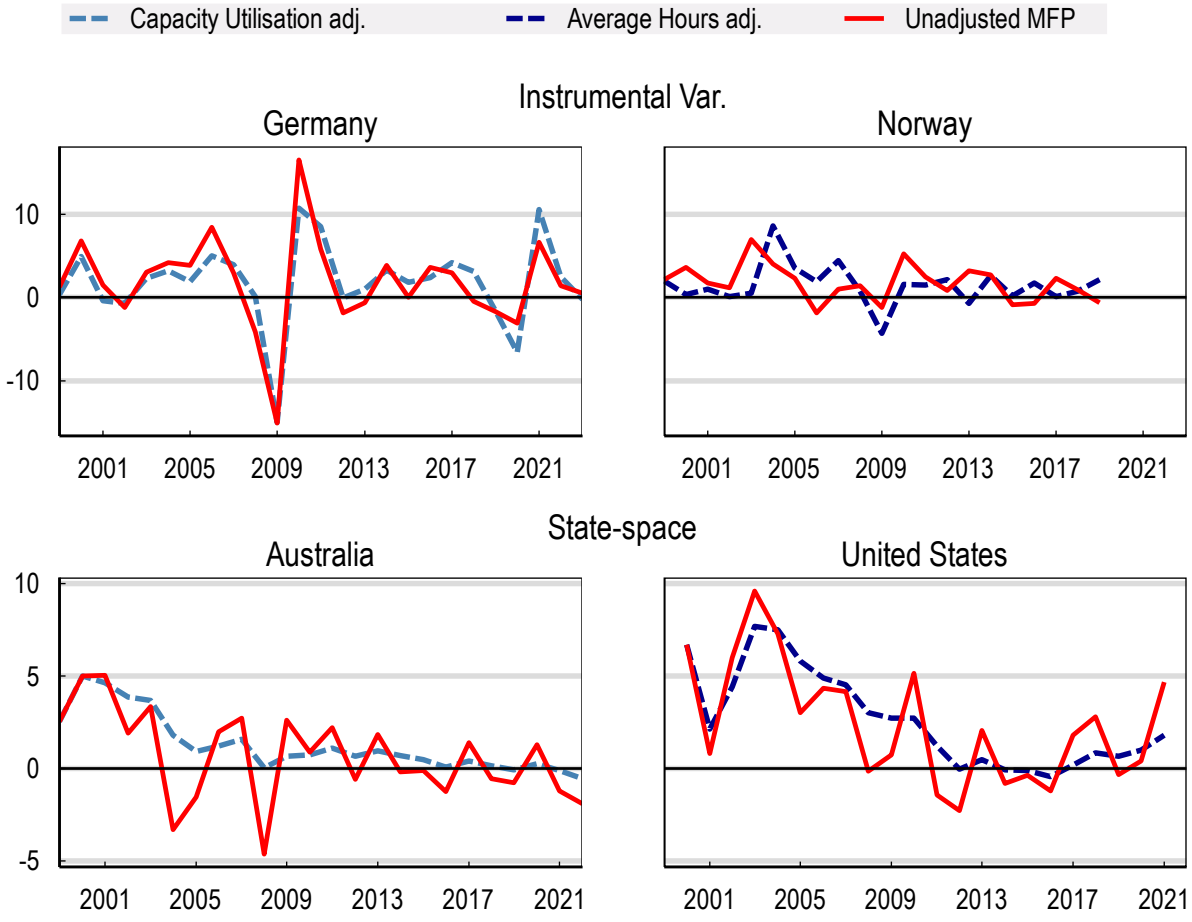
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What does cyclical adjustment reveal about long-term trends in MFP growth?

Adjusted MFP data reveal diverse long-term productivity trends across major economies (Figure 8.3). In Europe, adjusted MFP growth slowed from 2.4-3.7% in 2005 to between 0.5% and 2.2% in 2019. This masks significant internal divergence: Germany's adjusted MFP growth fell sharply from 1.7% in 2005 to -1.6% in 2019, whereas Italy maintained a consistently low level, barely shifting from 0.1% to 0% over the same period. Lithuania, by contrast, experienced sustained strong growth, only moderating slightly from 4.5-4.9% in 2005 to 3-4% in 2019.

The United States showed a clear downward trend, though adjusted MFP growth remained positive and above most Western European countries, declining from 3-5.8% in 2005 to 0.7-1.3% in 2019. Similarly, Japan experienced a decline in adjusted MFP growth, falling from around 2% in 2005 to -0.8% in 2019.

Figure 8.3. Adjusted MFP time series for selected OECD countries, 1995-2022



Note: The dark blue lines represent cyclically adjusted MFP growth based on capacity utilisation (CU) rates, while the lighter blue lines represent adjustments based on average hours worked per worker. The dashed lines indicate cyclical adjustments using instrumental variable (IV) regression, whereas the dot-dashed lines represent adjustments derived from a state-space model. The red line shows the unadjusted MFP series. The shorter IV regression series for Norway ends in 2020 due to data limitations in the instrumental variable for monetary policy shocks.
Source: (Arend, Botev and Jaax, forthcoming[6]).

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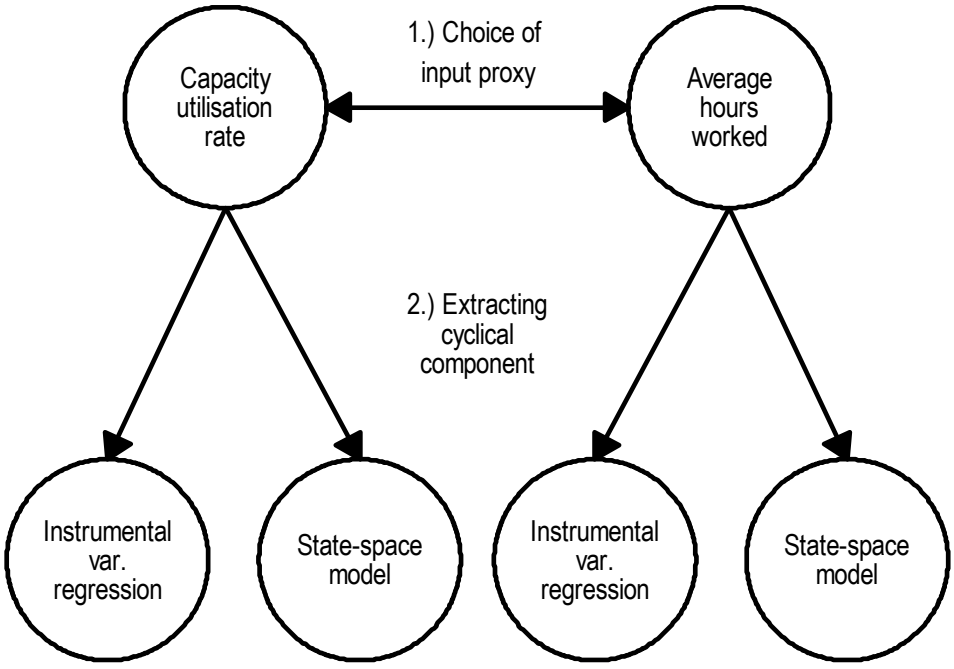
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Annex 8.A. Methodology

How to read the indicators

Cyclically-adjusted estimates of MFP growth require two design choices. First a proxy for the intensity of how inputs are used needs to be chosen. The literature has identified two main options: *average hours worked per worker* and self-reported *capacity utilisation rates from business surveys.* Second, the non-cyclical part of MFP growth needs to be extracted. This can be achieved either through *instrumental variable regressions* or through the use of *state-space models* and Kalman filters (Annex Figure 8.A.1). This chapter applied both approaches to gain insights into the evolution of cyclically-adjusted MFP growth in the manufacturing industry for 20 OECD countries for the time period 1995-2022.

Annex Figure 8.A.1. Main steps for the cyclical adjustment technique



Source: (Arend, Botev and Jaax, forthcoming[6]).

For the first step, while average hours worked per worker are frequently employed to adjust MFP, capacity utilisation rates from business surveys are seen to be more robust (Inklaar, 2006[7]; Martin and Jones, 2022[8]). Surveys provide a more direct measure of capacity utilisation than hours per worker as they are unaffected by changes in the composition of the workforce and changes in the relative cost of labour and capital.

For the second step, the IV regression approach is the most common in the literature (Comin et al., 2025[9]). The instruments chosen for the adjustment are country-specific oil price indices, economic policy uncertainty, monetary policy shocks and financial stress indicators. Overall, this approach performs well across the sample, with 15 out of 20 country-manufacturing panel IV regressions producing statistically significant results at the 90% confidence level.

In addition to the IV approach, a state-space framework to model the link between capacity utilisation surveys or average hours worked and the cyclical part of MFP growth was also employed. It allows estimating shifts and discrete breaks in MFP growth over time, which seems especially relevant in a period where MFP growth is slowing down and possibly affected by major economic shocks (Planas, Roeger and Rossi, 2013[10]). The model performed well for some countries for which the IV approach failed to provide statistically significant estimates (Annex Table 8.A.1).

Annex Table 8.A.1. Best cyclical-adjustment approach by country

Proxy / Modelling technique	Instrumental variable regression	State-space model
Capacity utilisation rates	AUT, CAN, DEU, EST, FIN, FRA, HRV, ITA, JPN, LTU, NLD, SVK, SVN,	AUS,
Average hours worked per worker	ESP, NOR	USA, GBR, LUX, ISR,

Note: The best approach here is determined by a combination of factors including statistical significance, strength of the instruments, residual correlation with GDP growth and visual analysis.
Source: (Arend, Botev and Jaax, forthcoming[6]).

Performance across models is robust

Both the instrumental variables approach and the state-space models each perform relatively well, in that they eliminate most of the correlation of MFP with GDP growth. However, they are highly dependent on specific assumptions and modelling choices.

Instrumental variable approaches primarily suffer from the challenge of identifying valid instruments. A good instrument must be strongly correlated with the cyclical component but uncorrelated with the underlying structural shocks or productivity innovations. In practice, finding such instruments can be difficult to find, and weak instruments can result in biased and inconsistent estimates as the results for the recent COVID-19 period demonstrate.

State-space models, while more flexible, rely on strong parametric assumptions, which may not align with the complex realities of productivity cycles. Misspecification of these assumptions or of the underlying state and measurement equations can seriously bias the estimated trend and cycle components.

9. Insights on productivity developments in 2024

Context and key findings

Productivity is central to economic policy. It underpins decisions on growth strategies, competitiveness and living standards. Yet official labour productivity data are published with a one- to two-year delay. Quarterly measures of GDP per hour worked can offer earlier insights, but they are not uniformly available across OECD and accession countries. When they exist, they are often subject to substantial revisions and sometimes provide only limited information on final annual estimates of productivity growth.

This chapter presents experimental nowcasts of annual labour productivity growth for 2024 across 40 OECD and accession countries. It builds on the approach of Dorville et al. (2025[1]), which offers improvements over standard alternatives in three respects. It is set in a panel setting, thus allowing to overcome the lack of data over a long time span, it considers a range of models, including dynamic factors models and machine-learning methods, and optimally exploit high frequency data through mixed-data sampling (Ghysels, Santa-Clara and Valkanov, 2004[2]; Borup, Rapach and Schütte, 2023[3]). The resulting estimates can provide policymakers with more timely insights into productivity dynamics, while highlighting areas of heightened uncertainty.

Key findings

- **Labour productivity growth is estimated to have remained weak across the OECD in 2024**, despite a modest rebound in Europe. The average estimated growth in labour productivity for OECD countries (excluding Türkiye) is 0.4%. The results suggest labour productivity gains in 2024 were modest in most regions: Europe (0.4%), Asia (0.2%), and North America (0.2%).
- **Labour productivity growth nowcasts varied widely across countries, with few clear leaders.** Only a handful of countries exhibited growth beyond 2.5% in 2024, including Croatia, Finland, Luxembourg and Chile. However, these figures come with sizeable uncertainty, especially Croatia and Chile, and should be interpreted cautiously.
- **The average labour productivity of G7 countries is estimated to have declined slightly (- 0.3%).** The nowcast points to a clear divergence between the United States – with estimated labour productivity growth of 1.5% – and the rest of the G7, which are expected to have experienced negative or near-zero productivity growth.
- **Nowcasts for 2024 reinforce the view that the post-pandemic productivity recovery remains fragile and uneven.** While estimates continue to converge towards historical averages, country-level nowcasts remain heterogenous and are surrounded by large uncertainties.

What happened in 2024?

Labour-productivity growth is expected to have remained weak in 2024. For the 40 OECD and accession countries included in the analysis, the nowcast points to an average increase of roughly 0.2 percentage points, a small improvement on the mild contraction recorded in 2023, but still far below pre-pandemic levels (Figure 9.1). For the G7 as a whole, the picture is mildly negative (-0.3%).

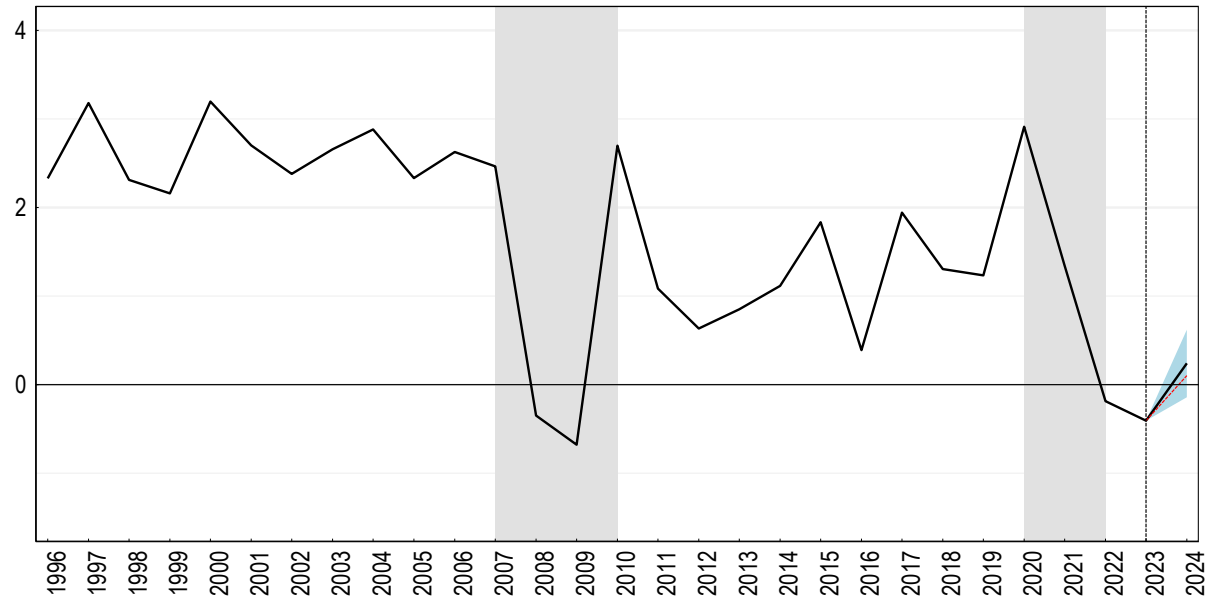
Taken together, the 2024 nowcasts reinforce the view that the post-pandemic productivity recovery remains fragile and uneven. Among regions, Europe appears closest to showing signs of a rebound (Figure 9.2). While the 26 European OECD countries are projected to have seen labour productivity growth by about 0.4%, this masks a pronounced split: strong gains in Finland, Luxembourg, and Croatia contrast with notable declines in large economies including Germany and Italy.

OECD countries outside Europe are estimated to have experienced only modest gains in labour productivity growth in 2024. Korea and Israel would have registered modest gains, whereas Japan would have slipped slightly into negative territory and Australia showed little change, leaving the regional average at only 0.2%. In 2024, North America is estimated to have exhibited a two-speed pattern, with solid productivity gains of about 1.5% in the United States contrasting with a small fall in Canada and a larger drop in Mexico, resulting in a regional mean of approximately 0.2%. South America would have experienced the strongest labour productivity growth at roughly 1.2%, primarily due to Chile. Yet, the region also exhibits the broadest confidence intervals, reflecting considerable uncertainty.

At the country level, dispersion is high (Figure 9.3). Twenty-three of the 40 OECD economies are projected to have seen positive labour productivity growth in 2024, but most of them by less than one percentage point. Only Croatia, Luxembourg, Finland and Chile break the 2.5% threshold, and all four come with sizeable uncertainties that call for caution when interpreting the numbers. Sixteen economies are expected to have experienced declines in labour productivity, led by Austria and Germany, while Ireland is projected to have experienced no change. Most point estimates fall within a fairly wide one percentage point confidence band.

Figure 9.1. Labour productivity growth nowcast for 2024

OECD average, Per cent

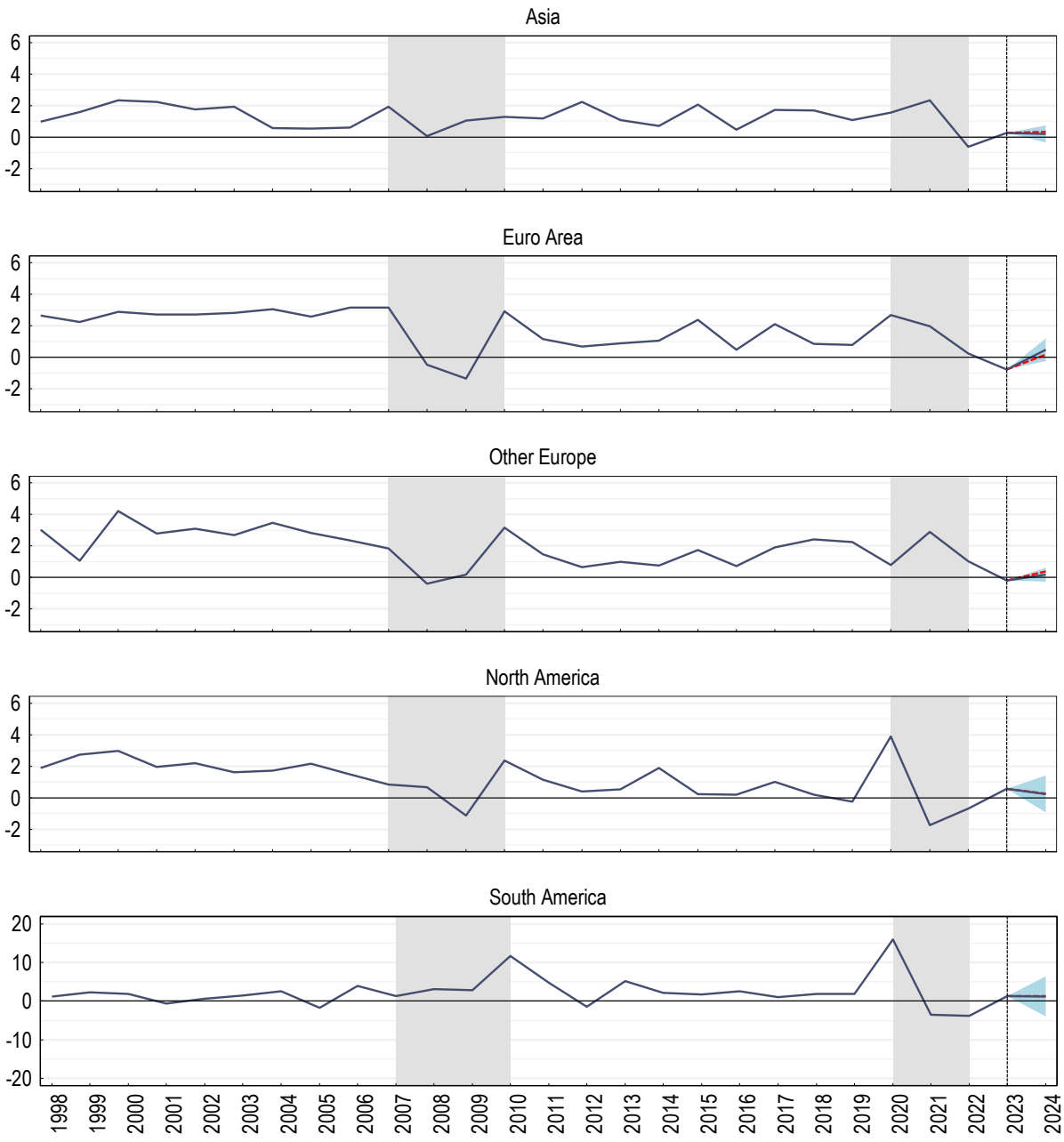


Note: The blue shaded area displays the 95% confidence intervals associated to the point estimate for the OECD aggregate. They are computed as $\left[\hat{\mu} \pm p_{97.5}\,\hat{\sigma}/\sqrt{N}\right]$ where $\hat{\mu}$ corresponds to the unweighted average prediction and $\hat{\sigma}$ the average Root Mean Squared Error. $p_{97.5} \approx 1.95$ is the 97.5th percentile of the normal distribution, and $N = 40$ the number of countries in the respective region. The red dotted line depicts the consensus nowcast, i.e., the average prediction of the five best models. Grey shaded areas represent global crisis episodes. Türkiye is excluded from the sample due to a large share of missing input data.
Source: Dorville et al. (2025[1]).

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Figure 9.2. Labour productivity growth by region, nowcast for 2024

Region-specific unweighted average of labour productivity growth, Per cent



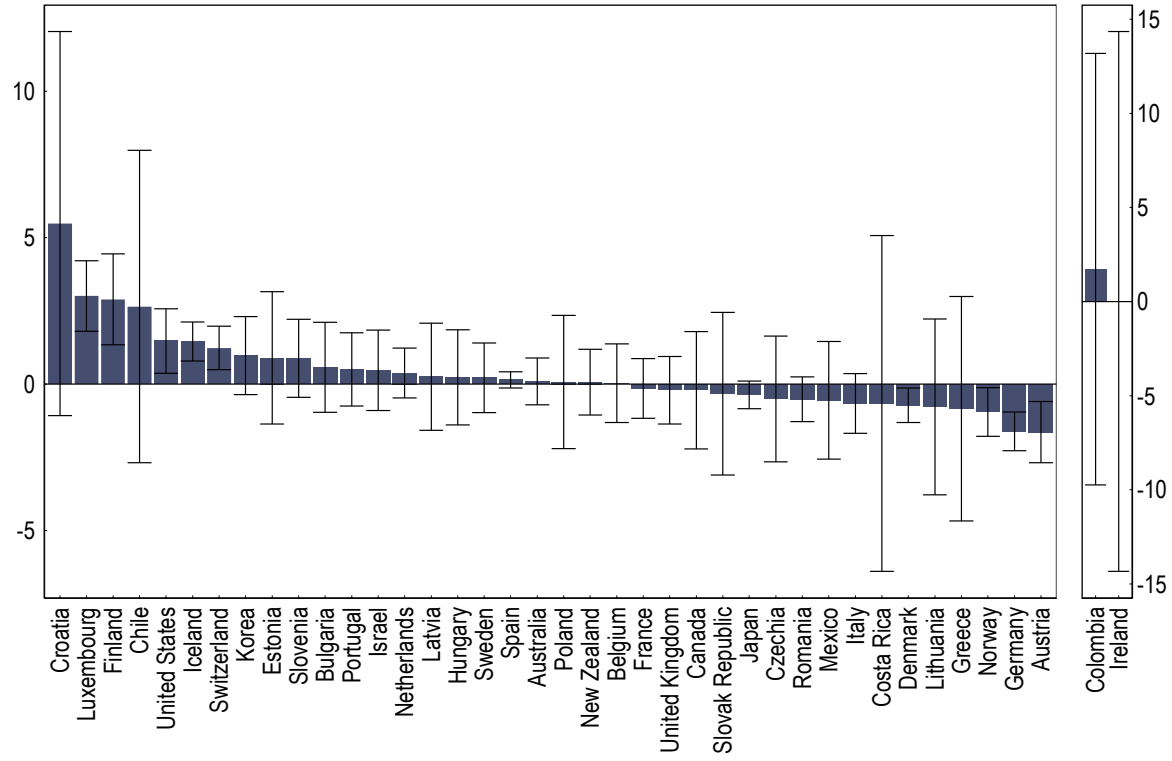
Note: The grey area refers to the 95% confidence bands associated with each predicted point. They are computed as $\left[\hat{\mu} \pm p_{97.5}\,\hat{\sigma}/\sqrt{N}\right]$ where $\hat{\mu}$ corresponds to the unweighted average prediction and $\hat{\sigma}$ the average RMSE, $p_{97.5} \approx 1.95$ is the 97.5th percentile of the standard normal distribution, and N the number of countries in the respective region. Red dotted lines correspond to the consensus prediction. Asia: Korea, Japan, Australia and New Zealand. Euro area: Austria, Belgium, Croatia, Estonia, Finland, France, Germany, Greece, Italy, Ireland, Latvia, Lithuania, Luxembourg, the Netherlands, Portugal, Slovak Republic, Slovenia and Spain. Other Europe: Denmark, Bulgaria, Romania, Switzerland, Hungary, Iceland, Poland, Sweden and Norway. North America: United States, Canada and Mexico, South America: Chile, Colombia and Costa Rica.

Source: Dorville et al. (2025[1]).

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Figure 9.3. Country-level labour productivity growth, nowcast for 2024

Predicted labour productivity growth per country, Per cent



Note: Vertical lines refer to the 90% confidence intervals associated with each predicted point.
Source: Dorville et al. (2025[1]).

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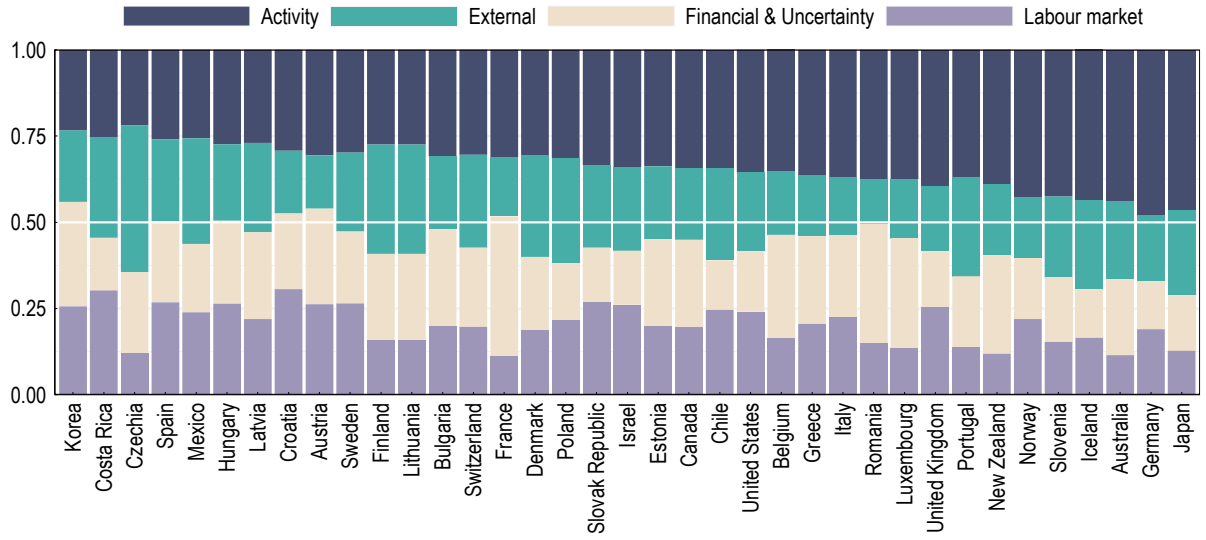
The underlying forces driving the predictions

Feature importance statistics reflect how much the model depends on each predictor to generate the nowcasts. Specifically, they quantify how much each predictor improves the accuracy of the nowcast on average.

First, activity-related indicators, such as those tracking industrial production and capacity utilisation, emerge as the most influential in driving labour productivity growth nowcasts across most countries, especially in advanced economies (Figure 9.4). The external environment also plays a role: external indicators (e.g. trade and foreign orders) are important in smaller, open economies like Czechia and Lithuania, while financial and uncertainty variables contribute more substantially in emerging or volatile contexts (e.g. Romania) or financial hubs like Luxembourg. The nowcast models for some countries (e.g. France, Canada, Greece) show a heavy reliance on global variables, indicating strong sensitivity to worldwide economic trends, while others (e.g. Switzerland, Slovak Republic, Iceland) rely more on country-specific indicators (Figure 9.5). In contrast, labour market indicators remain broadly secondary but can play a role, particularly in transition economies where changes in hours worked and employment exert greater influence.

Figure 9.4. Feature importance by domains

Average feature importance by domains

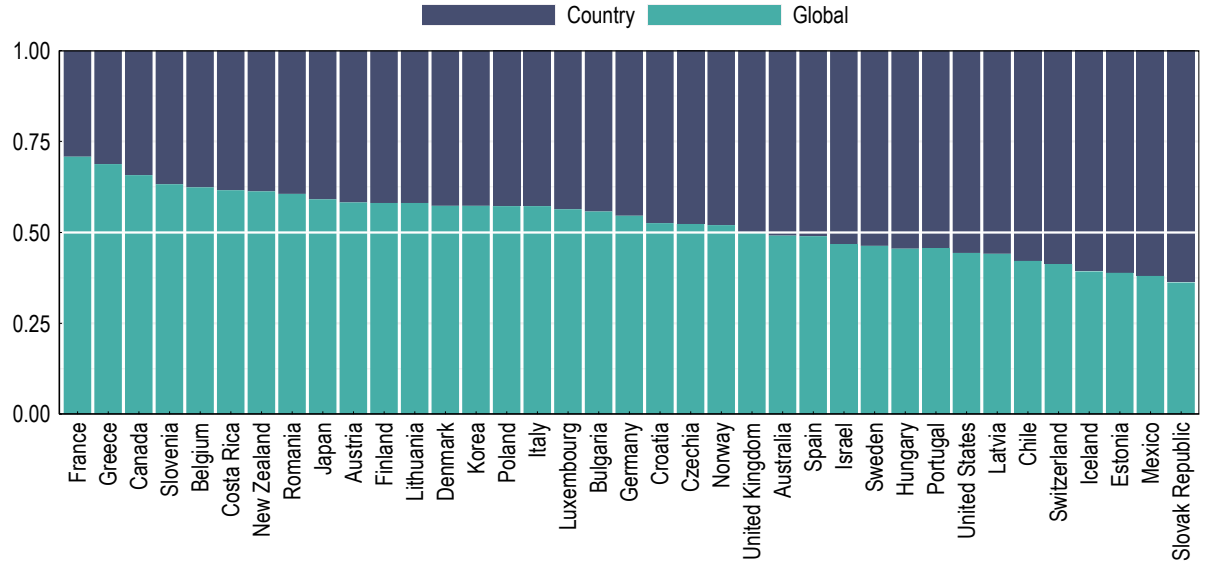


Note: Feature importance is measured as the average decrease in prediction error allowed by each feature. The resulting values are then normalised. Values are obtained from country-specific model of the imposed-weights MIDAS method and calculated over the period 2015-23.
Source: Dorville et al. (2025[1]).

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Figure 9.5. Feature importance by variable scope

Average feature importance by country-specific and global features



Note: Feature importance is measured as the average decrease in prediction error allowed by each feature. The resulting values are then normalised. Values are obtained from country-specific model of the imposed-weights MIDAS method and calculated over the period 2015-23.
Source: Dorville et al. (2025[1]).

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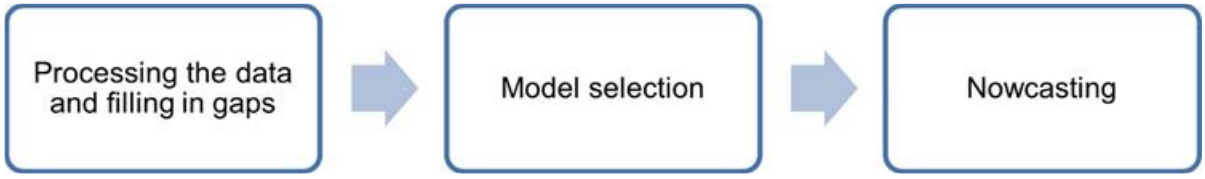
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Annex 9.A. Methodology

This annex provides a short overview of the approach to nowcast labour productivity growth, measured in terms of total hours worked, for 40 OECD and accession countries. A key specificity of the approach is to run models in a panel setting. As the target variable is annual and only available for the years 1995-2022, a panel approach increases the robustness of results (Woloszko, 2020[4]; Fosten and Nandi, 2023[5]; Mourougane et al., 2023[6]). In addition, models are estimated in a quasi real time accounting for publication lags, identified using lags observed in 2023.

The main steps are summarised in Annex Figure 9.A.1.

Annex Figure 9.A.1. Main steps of the empirical strategy



Source: OECD illustration.

Data

The dataset spans national-accounts aggregates, labour-market statistics, trade and business-cycle proxies, as well as a range of financial, geopolitical and policy-uncertainty measures for forty OECD countries. Because many of these series are not published in real time, the most recent observations are "now-filled": monthly indicators are prolonged with ARIMA projections, while higher-level series are extended with dedicated machine-learning models. The result is a balanced, country-by-frequency panel that replicates the information set that would have been available to forecasters at each historical point in time.

Model selection

Labour-productivity growth, the target variable, is annual and observed only for 1995-2022. Estimating models in a country panel therefore increases the effective sample size and exploits cross-sectional variation, an approach that has proved advantageous in earlier work by Woloszko (2020[4]), Fosten and Nandi (2023[5]) and Mourougane et al. (2023[6]). Publication delays are handled in quasi real time by imposing the calendar lags that prevailed in 2023 on every historical vintage. High-frequency information is folded into the annual horizon through Mixed-Data-Sampling regressions (Ghysels, Santa-Clara and Valkanov, 2004[2]). Three MIDAS variants are considered: an unrestricted version in which high frequency variables are included with no transformation; a restricted version in which high frequency variables follow a parsimonious lag polynomial; and an imposed-weight specification that aggregate high frequency variables with an equal weight, the later can be interpreted as a special case of the restricted model.

On top of the MIDAS bridge, six forecasting algorithms are explored: a dynamic-factor model that summarises co-movements across the data panel, penalised linear regressions (LASSO, Ridge and Elastic Net) that offer automatic variable selection; and tree-based machine-learning techniques, random forests and gradient-boosted trees, that capture non-linearities and interactions. Combining each algorithm with the three MIDAS lag-structures yields eighteen candidate specifications; an AR(1) in annual growth serves as a traditional benchmark, bringing the total to nineteen. All machine-learning hyper-parameters are tuned by grid search.

Performance is evaluated on one-year-ahead forecasts with the Root Mean Squared Error (RMSE). Rather than imposing the same specification on every economy, a single best-performing model is selected for each country. To guard against in-sample over-fitting, the evaluation follows an expanding-window cross-validation that starts in 2015 and moves forward one year at a time. For restricted MIDAS regressions, the polynomial lag structure and the regression coefficients are jointly estimated with the profile-likelihood technique of Ghysels and Qian (2019[7]), which retains estimation efficiency despite the added non-linearity.

Nowcasting

The selected country-specific models are applied to generate nowcasts for 2024. In parallel, a “consensus” estimate, the simple average of the five best individual models, is computed following the evidence that such ensembles reduce idiosyncratic model noise (Kotchoni, Leroux and Stevanovic, 2019[8]; Jaax, Mourougane and Gonzales, 2023[9]). Uncertainty is quantified with symmetric confidence bands whose half-width equals one model-specific RMSE, interpreted as the forecast-error standard deviation. Finally, for tree-based models, feature importance, measured as the decrease in squared error averaged over nodes, provides insight on the features underlying the nowcasts.

OECD Compendium of Productivity Indicators 2025

This report presents a comprehensive overview of both recent and long-term trends in productivity levels and growth across OECD countries and, where possible, in accession countries. It covers a range of topics, including recent developments in labour productivity, economic growth, sectoral reallocation, and multifactor productivity – both at the industry level and adjusted for cyclical effects. The report also examines trends in investment, labour productivity by firm size, and presents experimental estimates of productivity developments in 2024 for all 38 OECD member countries.

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